NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND
MIL-PRF-123B
6 August 1990
SUPERSEDING
MIL-C-123A
10 August 1982

PERFORMANCE SPECIFICATION

CAPACITORS, FIXED, CERAMIC DIELECTRIC,
(TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for high reliability, general purpose (BX and BR) and temperature stable (BP and BG) ceramic dielectric fixed capacitors, leaded and nonleaded for space, missile, and other high reliability applications. Capacitors covered by this specification may be used in critical frequency determining applications, timing circuits, and other applications where absolute stability is required (BP and BG), and in applications where appreciable variations in capacitance with respect to temperature, voltage, frequency, and life can be tolerated (BX and BR). Life tests in this specification are performed at two times rated voltage at maximum rated temperature, and an assumed acceleration factor of 8:1 is used to relate life test data obtained at two times rated voltage to performance at rated voltage.

1.2 Classification.

1.2.1 Military part number. Capacitors specified herein (see 3.1) shall be identified by a military part number which shall consist of the basic number of the military specification sheet followed by a series of coded characters. Each military specification sheet covers a different capacitor style. The coded number shall provide information concerning the capacitors' characteristic, working voltage, capacitance value, capacitance tolerance, and termination material. The military part number shall be in the following form as indicated:

M123	A	01	BX	B	103	K	C
Military specification number	Modification (see 1.2.1.1)	Slash sheet number	Characteristic (see 1.2.1.2)	Voltage (see 1.2.1.3)	Capacitance value (see 1.2.1.4)	Capacitance tolerance (see 1.2.1.5)	Termination (see 1.2.1.6)

1.2.1.1 Modification. The modification letter is used to indicate the latest characteristic of the part in the specification sheet. This letter shall be changed under direction of the preparing activity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, 3990 East Broad Street, Columbus, OH 43216-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

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1.2.1.2 Characteristic. The characteristic refers to the voltage-temperature limits of the capacitor. The first letter (B) identifies the rated temperature range of -55°C to +125°C. The second letter indicates the voltage-temperature limits as shown in table I.

TABLE I. Voltage-temperature limits.

Symbol	Capacitance change with reference to +25°C	
	Steps A through D of table XVI	Steps E through G of table XVI
G	90 ±20 ppm/°C	90 ±20 ppm/°C
P	0 ±30 ppm/°C	0 ±30 ppm/°C
R	+15, -15 percent	+15, -40 percent
X	+15, -15 percent	+15, -25 percent

1.2.1.3 Voltage. The rated voltage for continuous operation at +125°C is identified by a single letter as shown in table II.

TABLE II. Rated voltage.

Symbol	Rated voltage
	Volts, dc
A	25
B	50
C	100
D	200
E	500
K	150
L	300
M	400

1.2.1.4 Capacitance. The nominal capacitance value, expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When the nominal value is less than 10 pF, the letter "R" shall be used to indicate the decimal point and the succeeding digit(s) of the group shall represent significant figure(s). For example, 1R0 indicates 1.0 pF; R75 indicates .75 pF; and 0R5 indicates 0.5 pF.

1.2.1.5 Capacitance tolerance. The capacitance tolerance is identified by a single letter as shown in table III.

TABLE III. Capacitance tolerance.

Symbol	Capacitance tolerance ±
B	0.1 pF
C	0.25 pF
D	0.5 pF
F	1 percent
G	2 percent
J	5 percent
K	10 percent
M	20 percent

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1.2.1.6 Termination. The termination is identified by a single letter as shown in table IV.

TABLE IV. Termination.

Leaded capacitors		Nonleaded capacitors	
Symbol	Termination type	Symbol	Termination type
A	Copper-iron-zinc alloy (194 alloy)	G	Silver - nickel - gold
C	Copper, solder coated, 60 microinches, minimum	M	Palladium/silver alloy
W	Copper clad steel, solder coated, 60 microinches minimum	R	Palladium/silver alloy, solder coated
		S	Guarded, solder coated 1/
		T	Guarded, SN62 solder alloy (hot dipped)
		W	Base metallization - barrier metal - tinned (tin or tin-lead alloy)

1/ A guarded termination is one in which the interelectrode metallization is separated from the final finish by a barrier or guard metal; e.g., copper or nickel, etc.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Tin Alloy, Lead-Tin Alloy, and Lead Alloy.

MILITARY

MIL-C-39028 - Capacitors, Packaging of.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-1276 - Leads for Electronic Component Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION

RS-469 - Standard Test Method for Destructive Physical Analysis of High Reliability Ceramic Monolithic Capacitors.
RS-483 - Standard Method of Test for Effective Series Resistance (ESR) and Capacitance of Multilayer Ceramic Capacitors at High Frequencies.

(Application for copies should be addressed to the Electronic Industries Association (EIA), 1722 Eye Street, N.W., Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Capacitors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3).

3.2.1 Category A distributors. Category A distributors may be used under the following conditions:

a. The distributor shall be an authorized distributor of the manufacturer in accordance with MIL-STD-790.

b. The distributor shall not open the level of packaging containing the unit containers. This level of packaging shall bear a stamp stating "Not to be opened by the distributor." The distributor shall not remark or retest the product.

c. MIL-C-123 product shall be handled as a nonstocking item only.

d. The distributor shall be held accountable by the manufacturer to supply the complete data package to the user with each order.

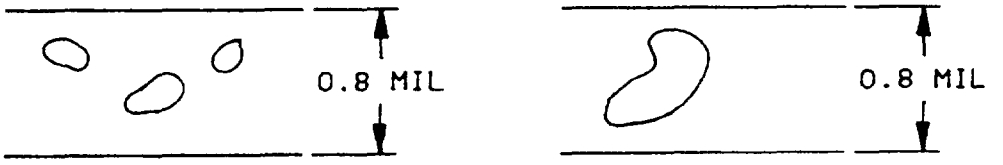
e. The manufacturer shall be responsible for auditing the distributors to the requirements of MIL-STD-790 and the limitations and controls as stated herein.

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3.3 Reliability and baseline documentation. Capacitors supplied to this specification shall conform to the requirements of MIL-STD-790 and the requirements specified herein. Prior to granting of qualification approval, a baseline document shall be prepared by the manufacturer and submitted to the qualifying activity for approval.

3.4 Design and construction. Capacitors shall be of the materials, design, construction, and physical dimensions specified (see 3.1) and in accordance with baseline documentation (see 3.3).

3.4.1 Dielectric parameters. Capacitors supplied to this specification shall have a minimum dielectric thickness of 0.8 mils and a maximum dielectric constant of 2400 (see figure 1). Dielectric thickness is the actual measured thickness of the dielectric layer. Voids, or the cumulative effect of voids, shall not reduce the total dielectric thickness by more than 50 percent.



Acceptable Unacceptable

FIGURE 1. Dielectric parameters.

3.4.2 Lead attachment (leaded capacitors only) (see 3.1). Capacitors with leads shall have the leads attached to the ceramic chip body using high temperature solder with a minimum plastic point of +260°C.

3.4.3 Lead wire. Lead wire shall be in accordance with MIL-STD-1276 (see 3.1).

3.5 Nondestructive internal examination (see appendix A). Prior to termination, all capacitors delivered to this specification shall be subjected to neutron radiography, ultrasonic examination or some other method approved by the baseline (see 3.3). The method of examination shall be determined by the manufacturer. N-ray shall not be performed on characteristics BP and BG capacitors.

3.5.1 Neutron radiography (N-ray). When capacitors are tested as specified in 4.6.1.1, there shall be no devices delivered that exhibit defects defined in appendix A.

3.5.2 Ultrasonic examination. When capacitors are tested as specified in 4.6.1.2, there shall be no devices delivered that show unacceptable responses as defined in appendix A.

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3.6 Pretermination destructive physical analysis. When examined as specified in 4.6.2, parts shall meet the requirements of EIA standard RS-469.

3.7 Visual examination.

3.7.1 Pre-encapsulation visual inspection (leaded capacitors only). When leaded capacitors are examined as specified in 4.6.3, they shall meet the visual inspection requirements specified in appendix B.

3.7.2 Visual examination of nonleaded capacitors. When examined as specified in 4.6.3, capacitors shall meet the visual requirements specified in appendix C.

3.8 Pre-encapsulation terminal strength (leaded capacitors only). Capacitors shall be tested as specified in 4.6.4 and appendix D. The force shall be applied in the manner specified until rupture and the level recorded. In no case shall the termination rupture at a load less than that specified in table XIV.

3.9 Radiographic inspection (encapsulated capacitors only). When radiographed in accordance with 4.6.5, leaded capacitors shall meet the following requirements (see appendix E):

a. There shall be no evidence of improperly made connections, substandard soldering, structural weakness, or attached solder particles or slivers.

b. The interface between each lead and capacitor element shall have solder for at least 80 percent of the interface dimension (see figure E-3).

c. The encapsulation material shall have a minimum thickness of .005 inch (0.13 mm) (dimension T, figure E-3), with the exception of the lead side of the radial leaded capacitors, which shall have a minimum thickness of .010 inch (0.25 mm) (dimensions T_R, figure E-3).

d. If a lead has a portion which has been flattened to accommodate the ceramic element more easily, the formed portion of the lead shall be completely inside the encapsulation material. Any unintentional nick or deformation of the leads shall not reduce the lead diameter below 90 percent of the minimum diameter required (see 3.1).

e. The encapsulation material thickness shall be at least .005 inch (0.13 mm) on all external surfaces of the chip and any protrusions of termination, solder, or lead frame.

f. Extraneous particles shall be no larger than .005 inch (0.13 mm) in any dimension. Voids in the encapsulant shall not reduce the thickness below the minimum value.

3.10 Thermal shock and voltage conditioning. When tested as specified in 4.6.6, capacitors shall withstand the extremes of high and low temperature without visible damage and meet the following requirements with the percent defective allowed as shown in table XV:

a. Dielectric withstanding voltage (at +25°C): As specified in 3.13.

b. Insulation resistance (at +25°C): Shall be not less than the initial requirement.

c. Insulation resistance (at elevated ambient temperature): Shall be not less than the initial requirement (see 3.14).

d. Dissipation factor (at +25°C): Shall not exceed the value specifed (see 3.12).

e. Capacitance (at +25°C): Shall be within the tolerance specified (see 3.1).

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Capacitance values of no more than 5 percent or .5 pF, whichever is greater, for BX and BR characteristics or of no more than 1 percent or .3 pF, whichever is greater, for BP and BG characteristics beyond specified tolerance limit shall be removed from the lot but shall not be considered defective for determination of the percent defective allowed.

3.11 Capacitance (Cap). When measured as specified in 4.6.7, the capacitance shall be within the specified tolerance (see 3.1).

3.12 Dissipation factor (DF). When determined as specified in 4.6.8, the dissipation factor shall not exceed 2.5 percent for BX and BR, 0.15 percent for BP, and 0.01 percent for BG, unless otherwise specified (see 3.1).

3.13 Dielectric withstanding voltage (DWV). Capacitors shall withstand the direct current (dc) potential specified in 4.6.9 without damage or breakdown.

3.14 Insulation resistance (IR). When measured as specified in 4.6.10, the insulation resistance shall be not less than the value specified:

a. At +25°C: 100,000 megohms or 1,000 megohm-microfarads, whichever is less.

b. At +125°C: 10,000 megohms or 100 megohm-microfarads, whichever is less.

3.15 Destructive physical analysis. When examined as specified in 4.6.11, capacitors shall meet the requirements of EIA standard RS-469.

3.16 Terminal strength. When capacitors are tested as specified in 4.6.12, there shall be no loosening or rupturing of the terminals.

3.17 Solderability.

3.17.1 Solderability (leaded capacitors). When leaded capacitors are tested as specified in 4.6.13.1, the dipped surface of the leads shall be at least 90 percent covered with a new, smooth, solder coating. The remaining 10 percent may contain only small pinholes or rough spots; these shall not be concentrated in one area. Bare base metal where the solder dip failed to cover the original coating is an indication of poor solderability, and shall be cause for failure. In case of dispute, the percent of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area.

3.17.2 Solderability (nonleaded capacitors). Capacitors shall be tested as specified in 4.6.13.2, except Sn62, in accordance with QQ-S-571, shall be used in lieu of Sn60 and the immersed metallized surface shall be at least 85 percent covered with a smooth solder coating. The remaining 15 percent of the surface may contain small pinholes or exposed termination material; however, these shall not be concentrated in one area.

3.18 Resistance to soldering heat.

3.18.1 Resistance to soldering heat (leaded capacitors). When tested as specified in 4.6.14.1, capacitors shall meet the following requirements:

a. Insulation resistance at +25°C: Not less than the initial +25°C requirement (see 3.14).

b. Capacitance change:

(1) BR and BX characteristics: Shall change not more than -1.0 percent to +6.0 percent from the initial measured value.

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(2) BG and BP characteristics: Shall change not more than -1.0 percent to +2.0 percent or 0.5 pF, whichever is greater, from the initial measured value.

c. Dissipation factor: Shall not exceed the initial limits (see 3.12).

3.18.2 Resistance to soldering heat (nonleaded capacitors). When tested as specified in 4.6.14.2, capacitors shall meet the following requirements:

a. Visual examination: There shall be no evidence of mechanical damage or delamination or exposed electrodes. Leaching shall be a maximum of 25 percent on each edge of mounting area (see figure 2).

b. Insulation resistance, capacitance change, and dissipation factor shall be measured as specified in 3.18.1.

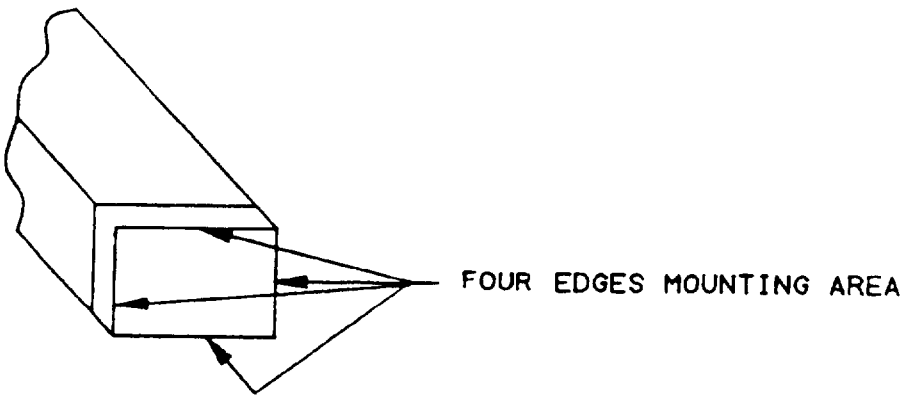


FIGURE 2. Mounting area.

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3.19 Voltage-temperature limits. The capacitance change over the range of temperatures specified in 4.6.15 shall not exceed the limits specified in table V. The capacitance value obtained in step C of table XVI shall be considered as the reference point. For BP and BG values below 20 pF, see table VI.

TABLE V. Capacitance change with reference to +25°C.

Characteristic	Steps A through D of table XVI Bias = 0 volt	Steps E through G of table XVI Bias = rated voltage
BR	±15 percent	+15, -40 percent
BX	±15 percent	+15, -25 percent
BP 1/	0 ±30 ppm/°C	0 ±30 ppm/°C
BG 1/	90 ±20 ppm/°C	90 ±20 ppm/°C

1/ At the +25°C stability point, the capacitance measurement may be ±.1 percent or 0.05 pF, whichever is greater, from the +25°C reference.

TABLE VI. Temperature coefficient tolerances for characteristics BP and BG values below 20 pF.

Permissible capacitance change from capacitance at +25°C in ppm/°C					
Temperature	Less than 2.1 pF	2.15 pF to 4.2 pF	4.3 pF to 8.0 pF	8.1 pF to 18 pF	Greater than 18 pF
+125°C	1/	±250 ppm/°C	±120 ppm/°C	±60 ppm/°C	±30 ppm/°C
-55°C 2/	1/	+246.25 -326.25	+116.25 -166.25	+55.00 -91.25	+27.50 -53.75

1/ Not practically measurable.
2/ The ppm/°C values for -55°C were calculated by dividing ppm by -80°C.

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3.20 Moisture resistance. When tested as specified in 4.6.16, capacitors shall meet the following requirements:

a. Visual examination: No mechanical damage. Marking shall remain legible.

b. Dielectric withstanding voltage (moisture resistance, normal voltage only): As specified in 3.13.

c. Insulation resistance: Humidity, steady state, 1.3 ±0.25 volts - Shall meet the initial +25°C requirement specified in 3.14. Moisture resistance, normal voltage - Not less than 50 percent of the initial +25°C requirement specified in 3.14.

d. Capacitance: Change not to exceed ±10 percent from initial measured value for BX and BR characteristics, and not more than ±.3 percent or .3 pF, whichever is greater, for BP and BG characteristics.

3.21 Vibration, high frequency (leaded capacitors only). When capacitors are tested as specified in 4.6.17, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, open or short circuitry, or evidence of mechanical damage.

3.22 Resistance to solvents. When capacitors are tested as specified in 4.6.18, there shall be no evidence of mechanical damage and the marking shall remain legible.

3.23 Life (at elevated ambient temperature). When tested as specified in 4.6.19, capacitors shall meet the following requirements:

a. 250-hour limits:

(1) Insulation resistance (at +125°C): Shall not be less than 50 percent of the value specified (see 3.14).

(2) Visual examination: No mechanical damage. Marking shall remain legible.

(3) Insulation resistance (at +25°C): Shall not be less than 50 percent of the value specified (see 3.14).

(4) Capacitance: Change not to exceed ±15 percent from the initial measured value for BX and BR characteristics, and ±0.3 percent or 0.3 pF, whichever is greater, from the initial measured value for BP and BG characteristics.

(5) Dissipation factor: Shall not exceed the value specified (see 3.12).

(6) Equivalent series resistance (when specified, see 3.1): Shall meet the requirements specified in 3.24 and 3.25.

b. 1,000-, 2,000-, and 4,000-hour limits:

(1) Insulation resistance (at +125°C): Shall not be less than 30 percent of the value specified (see 3.14).

(2) Visual examination: No mechanical damage. Marking shall remain legible.

(3) Insulation resistance (at +25°C): Shall not be less than 30 percent of the value specified (see 3.14).

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(4) Capacitance: Change not to exceed ±20 percent from the initial measured value for BX and BR characteristics, and ±0.5 percent or 0.5 pF, whichever is greater, from the initial measured value for BP and BG characteristics.

(5) Dissipation factor: Shall not exceed 3.0 percent for BX and BR characteristics, and 0.15 percent for BP and BG characteristics.

(6) Equivalent series resistance (when specified, see 3.1): Shall meet the requirements specified in 3.24 and 3.25.

3.24 Equivalent series resistance (UHF) (when specified, see 3.1). When tested in accordance with 4.6.20, the ESR shall be less than the limits shown in appendix F, figures F-1 and F-2.

3.25 Equivalent series resistance (RF) (when specified, see 3.1). When tested in accordance with 4.6.21, the ESR shall be less than the limits shown in appendix F, figures F-1 and F-2.

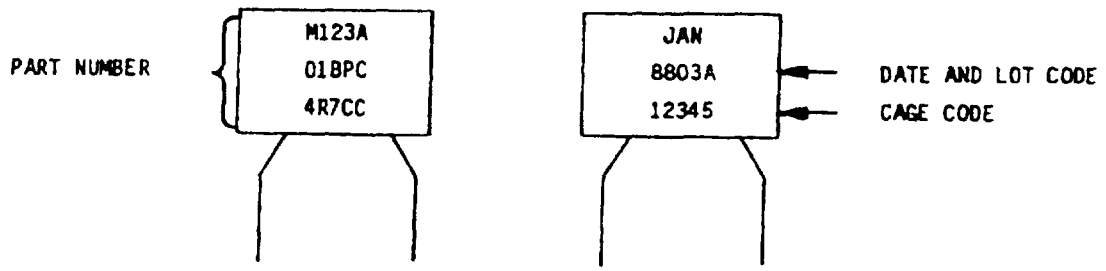
3.26 Resonance frequency (when specified, see 3.1). When tested as specified in 4.6.22, capacitors shall meet or exceed the resonance frequency as shown in appendix F, figure F-3.

3.27 Marking capacitors. Capacitors shall be marked as specified herein. Paper labels shall not be used. Other markings which in any way interfere with, obscure, or confuse those specified herein, are prohibited. Each capacitor shall be legibly marked and will withstand the environmental conditions specified herein. The marking shall be a minimum resistance of 1×10^9 ohms per square and shall remain legible after coating the parts with 1.5 to 3 mils of transparent conformal coating such as polyurethane, acrylic, or the equivalent.

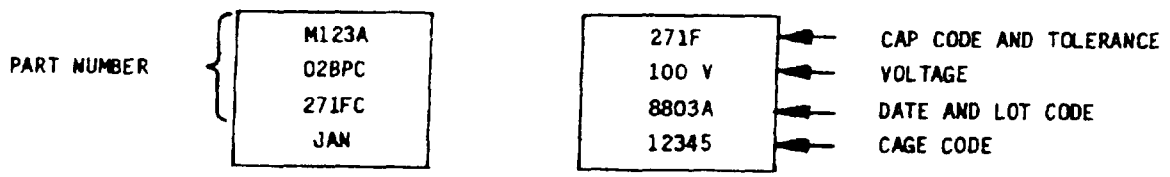
3.27.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control, over the certification marks "JAN" and "J", respectively, to indicate electrical equipment, namely, resistors, capacitors, electron tubes and the like, acquired by, or manufactured for use by, or for the Government in accordance with standard Government specifications. Accordingly, capacitors acquired to, and meeting all of the criteria specified herein and in applicable specification sheets shall bear the certification mark "JAN", except that capacitors too small to bear the certification mark "JAN" shall bear the letter "J". Capacitors furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable specification sheets shall not bear "JAN" or "J". In the event a capacitor sample fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all capacitors represented by the sample. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN". Unleaded capacitors to this specificaiton shall not be marked with "JAN" or "J"; however, the packaging containers shall bear "JAN" or "J".

3.27.2 Full marking (leaded capacitors). Unless otherwise specified (see 3.1), capacitors shall be marked with the "JAN" or "J" marking, part number, date and lot code, manufacturer's five digit commercial and government entity (CAGE) code, voltage, capacitance, and capacitance tolerance. There shall be no space between the symbols whch comprise the part number. The date code and lot number shall consist of the year, week, and lot code. For example: The third week of 1990 would be 9003-. Capacitors shall be marked in accordance with one of the examples shown on figure 3 (see 3.1).

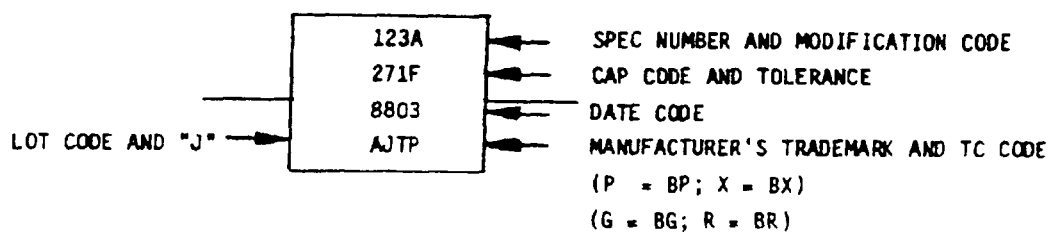
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Example 1



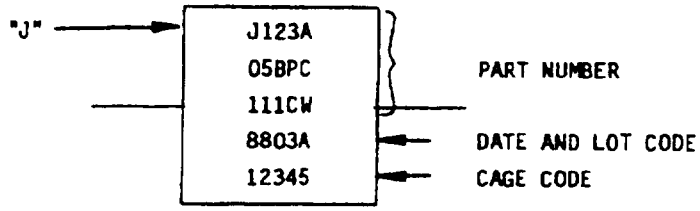
Example 2



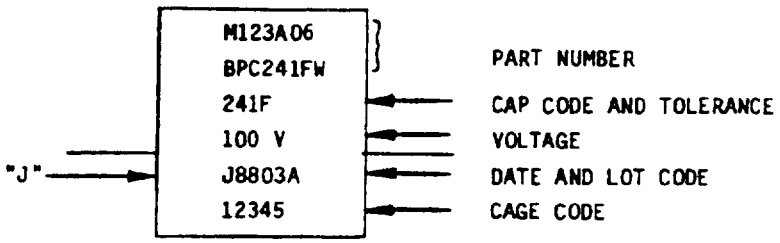
Example 3

FIGURE 3. Marking examples for leaded capacitors.

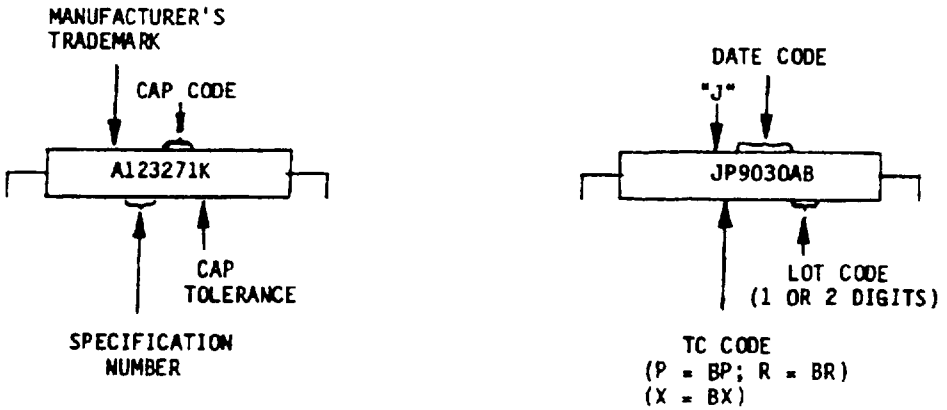
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Example 4



Example 5



Example 6

FIGURE 3. Marking examples for leaded capacitors - Continued.

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3.27.3 Marking (nonleaded capacitors). Capacitors shall be legibly marked in a contrasting color, with a three-digit capacitance code (see 1.2.1.4). When capacitors are too small for three digits, they shall be marked in accordance with table VII.

TABLE VII. Marking for small nonleaded capacitors.

Character	Significant units	Capacitance (pF) and multiplier					
		Orange (X 0.1)	Black (X 1.0)	Green (X 10)	Blue (X 100)	Violet (X 1000)	Red (X 10,000)
A	10	1.0	10	100	1000	10,000	100,000
B	11	1.1	11	110	1100	11,000	110,000
C	12	1.2	12	120	1200	12,000	120,000
D	13	1.3	13	130	1300	13,000	130,000
E	15	1.5	15	150	1500	15,000	150,000
H	16	1.6	16	160	1600	16,000	160,000
I	18	1.8	18	180	1800	18,000	180,000
J	20	2.0	20	200	2000	20,000	200,000
K	22	2.2	22	220	2200	22,000	220,000
L	24	2.4	24	240	2400	24,000	240,000
N	27	2.7	27	270	2700	27,000	270,000
O	30	3.0	30	300	3000	30,000	300,000
R	33	3.3	33	330	3300	33,000	330,000
S	36	3.6	36	360	3600	36,000	360,000
T	39	3.9	39	390	3900	39,000	390,000
V	43	4.3	43	430	4300	43,000	430,000
W	47	4.7	47	470	4700	47,000	470,000
X	51	5.1	51	510	5100	51,000	510,000
Y	56	5.6	56	560	5600	56,000	560,000
Z	62	6.2	62	620	6200	62,000	620,000
3	68	6.8	68	680	6800	68,000	680,000
4	75	7.5	75	750	7500	75,000	750,000
7	82	8.2	82	820	8200	82,000	820,000
9	91	9.1	91	910	9100	91,000	910.000

3.27.4 Marking (nonleaded high frequency capacitors). Capacitors shall be laser marked with manufacturer's trademark or symbol, the three-digit capacitance code and tolerance code. Alternately, a three-digit capacitance code and tolerance code in accordance with table VII may be used. Capacitance shall be marked parallel to the plane of the embedded electrodes.

3.28 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality when using 10X magnification. External leads shall not exhibit cuts, nicks, or scrapes exceeding 10 percent of the diameter for round and flat lead wire parts, except as follows:

a. Round leads: Within .050 inch of the body of the component, 10 percent of the surface area of the leads may exhibit bare base metal. These capacitors are not expected to be solderable within .050 inch of the case.

b. Flat leads (styles CKS22, CKS23, and CKS24): Surface of lead may exhibit bare base metal on edges except on installation portion of lead. These capacitor leads are not expected to be solderable above the plane of the body mounting base.

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3.29 Supplying to looser capacitance tolerance and lower rated voltage. Parts qualified and marked to tighter capacitance tolerance or higher rated voltage, with procuring agency approval, are substitutable for parts marked to looser capacitance tolerance or lower rated voltage, provided all other values, such as case size, characteristic, and leads are the same. The substitutable parts shall not be remarked unless specified in the contract or purchase order (see 6.2), the lot date codes on the parts are unchanged, and the workmanship criteria is met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. Procedures for product and quality audits are specified in appendix G.

4.1.3 Sample and data retention - all manufacturing lots. To facilitate failure analyses which may be required in cases of performance problems occurring in the service life of parts destined for use in long-term programs, the following samples and data relating to material, process, and product shall be retained by the vendor, and maintained in usable condition for 15 years from the date of shipment of the parts.

4.1.3.1 Material control documentation.

4.1.3.1.1 Purchased raw materials. The following documentation for purchased raw materials shall be retained:

a. Procurement documentation: Traceability.

b. Physical and chemical property data.

c. Performance evaluation/characterization data.

4.1.3.1.2 In-house prepared materials. The following documentation for in-house prepared materials shall be retained:

a. Fabrication process control data.

b. Physical and chemical property data.

c. Performance evaluation/characterization data.

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4.1.3.2 Process control documentation. The following process control documentation shall be retained:

a. Lot travelers, including material traceability.

b. In-process nondestructive test results; e.g., acoustic emission, NRI, X-ray film.

c. Process control: Critical parameter data.

4.1.3.3 In-process and finished product test samples and data. The following in-process and finished product test samples and data shall be retained:

a. In-process (preliminary) DPA samples and test report.

b. All manufacturing lots.

(1) Group A: Final DPA samples, electrical test samples, and test data.

(2) Group B: Test samples and test data.

c. Selected lots where group C testing is conducted: Test samples and test data.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. In-process inspection (see 4.5.1).

c. Quality conformance inspection (see 4.5.2 and 4.5.3).

4.3 Inspection conditions and reference measurements.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202, except relative humidity shall not exceed 75 percent. Accuracy of all test voltage measurements shall be within ±2.0 percent of the specified voltage.

4.3.2 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at +25°C ±3°C prior to conditioning. Unless reference measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.3 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the specified test voltage.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the qualifying agency (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number of capacitors to be specified for qualification inspection shall be as specified in table VIII and in appendix H of this specification. Each capacitor style shall be qualified separately.

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TABLE VIII. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures 1/
Group I				
Radiographic inspection (leaded devices only)	3.9	4.6.5	All	N/A
Thermal shock and voltage conditioning	3.10	4.6.6	186 min 2/ 3/	See table XV
Dielectric withstanding voltage	3.13	4.6.9		
Insulation resistance +25°C	3.14	4.6.10		
Insulation resistance +125°C	3.14	4.6.10		
Capacitance	3.11	4.6.7		
Dissipation factor	3.12	4.6.8		
Equivalent series resistance (UHF) (high frequency only)	3.24	4.6.20		
Equivalent series resistance (RF) (high frequency only)	3.25	4.6.21		
Group II				
Visual and mechanical examination; material, design, construction and workmanship	3.1, 3.4, 3.27, and 3.28	4.6.3	15	1
Destructive physical analysis	3.15	4.6.11		
Group IIIa - Leaded devices				
Terminal strength	3.16	4.6.12.1	12	1
Solderability	3.17.1	4.6.13.1		
Resistance to soldering heat	3.18.1	4.6.14.1		
Group IIIb - Nonleaded devices 4/				
Terminal strength	3.16	4.6.12.2	12	1
Solderability	3.17.2	4.6.13.2	12	
Resistance to soldering heat	3.18.2	4.6.14.2		
Group IV 5/				
Voltage-temperature limits	3.19	4.6.15	12	1
Moisture resistance	3.20	4.6.16.2		
Group V				
Humidity, steady state, low voltage	3.20	4.6.16.1	12	0
Vibration, high frequency (leaded capacitors only)	3.21	4.6.17	12	1
Resistance to solvents	3.22	4.6.18		

See footnotes at end of table.

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TABLE VIII. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures 1/
Group VI				
Life	3.23	4.6.19	123	1
Group VII				
Resonance frequency (high frequency only)	3.26	4.6.22	12	0

1/ A sample unit having one or more defects will be charged as a single defective.
2/ Additional samples over the 186 minimum should be included, based on table XV to allow for the percent defective allowable. Twelve additional samples shall be required for nonleaded devices.
3/ An additional 12 samples shall be required for high frequency devices.
4/ Or 24 high frequency devices.
5/ Leads may be soldered to chip capacitor to facilitate the tests required in group IV.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table VIII in the order shown. Sample units shall have been subjected to the in-process screening required by this specification and all sample units shall be subjected to the inspection of group I. The sample shall then be divided as specified in table VIII as applicable.

4.4.3 Failures. Failures in excess of those allowed in table VIII shall be cause for refusal to grant qualification approval.

4.4.4 Retention of qualification. To retain qualification for capacitors specified in this specification, the manufacturer shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall include the following:

a. A list of all device types for which the manufacturer has qualification, showing the total number of lots completing quality conformance inspection during the reporting period.

b. Certification that the manufacturer still has the capabilities and facilities necessary to produce these items to the qualified levels.

c. The number of lots which failed lot by lot testing or periodic testing (group B or group C).

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d. For failed lots (groups B and C), the following information is required:

(1) Inspection lot identification code and part number(s) in the failed lot.

(2) The percent defective of voltage conditioning.

(3) The results of groups A, B, and C testing (i.e., pass or fail).

(4) For failed subgroups, the number and mode of all failures.

(5) All lots withdrawn shall be counted as failed lots in the report.

(6) Identification of the lots (by inspection lot identification code, part number, and date of completion of group B testing).

e. Certification for each device style which has been qualified, stating that at least two lots have been manufactured utilizing the design and construction for which qualification is valid during the 12-month reporting period.

The report shall also include certification that the design and construction of the product was verified and found to be identical to the style for which qualification approval was valid. If the summary of test results indicates nonconformance with the requirements of this specification and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the summary report within 30 days after the end of 12-month period may result in loss of qualification for products within that capacitor's group. In addition to the periodic submission of inspection data, the manufacturer shall within 30 days notify the qualifying activity at any time during the 12-month period that the inspection data indicates inability of the qualified product to meet the requirements of this specification. If during two consecutive reporting periods there has been no quality conformance inspection of product within a qualified technology, the manufacturer may, at the discretion of the qualifying activity, be required to submit a representative product(s) of that qualified technology to complete testing in accordance with the qualification requirements. In this case the product submitted must comply with the baseline for each dielectric type.

4.5 In-process inspection and quality conformance inspection.

4.5.1 In-process inspection. Each production lot of parts shall be inspected in accordance with table IX. The use of neutron radiography or ultrasonic examination shall be as specifed in baseline documentation. Other screening examinations may be applied at the option of the manufacturer, as documented and approved by the baseline (see 3.3).

4.5.1.1 Rework. Rework is not allowed on any operations including and subsequent to the formation of the slurry, except for marking and lead rework.

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TABLE IX. In-process inspection.

Inspection	Requirement paragraph	Test method paragraph	Sample size
Nondestructive internal examination	3.5	4.6.1	100 percent
Pretermination destructive physical analysis	3.6	4.6.2	Table XIII
Visual examination	3.7	4.6.3	100 percent
Pre-encapsulation terminal strength (leaded capacitors only)	3.8	4.6.4	Table XIV

4.5.2 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.2.1 Inspection lot. An inspection/production lot shall be of one design as specified in the baseline documentation. As a minimum requirement, the lot shall consist of all the capacitors of a single nominal capacitance/voltage rating of one design, from the same dielectric material batch, and processed as a single lot through all manufacturing steps on the same equipment, to the same baseline/revision, and identified with the same date and lot code designation. The lot may contain all available capacitance tolerances for the nominal capacitance value (see baseline documentation for BP and BG characteristics). In addition, the lot shall conform to the following:

a. Raw materials, such as end terminations, ceramic raw material, binders, and electrode ink, shall be traceable to the same lot or batch and be from the same contractor.

b. A lot number shall be assigned before electrode printing (inking) and be manufactured from the same slurry and cast on the same belt or comparable equipment (see 3.4).

c. A single mix of a basic ceramic formulation fired through one kiln with one temperature profile during the process, with constant dielectric design, with, in general, no more than 20 percent variation in the number of active dielectrics being assembled in one continuous buildup. Sublots may be allowed to provide a spread in capacitance value to improve the yield. This will normally be done in the case of tight tolerances and low capacitance. In some cases, more than 20 percent variation in the number of active dielectrics may be required in these circumstances (see 3.3).

d. End termination material shall be consistent in formulation and traceable to a single batch and shall be fired in the same kiln with no temperature adjustment during the process. All parts shall enter the kiln in one continuous operation/run for end termination firing.

e. Lead attachment solder shall be of a uniform composition from a single supplier with a minimum melting point of +260°C.

f. Lot identity shall be maintained from the time the lot is assembled to the time it is accepted.

g. The lot date code (LDC) shall be added after firing and prior to end termination for leaded devices, and after end termination firing for nonleaded devices.

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4.5.2.2 Group A inspection. Group A inspection shall consist of the tests and examinations specified in table X. The following details shall apply:

a. Failure to meet destructive physical analysis (DPA) limits shall cause the lot to be rejected.

b. A summary of the results of group A inspection on each lot that meets requirements shall be submitted to the purchaser with the parts.

c. A copy of the applicable DPA report shall be submitted to the purchaser with each lot of parts.

TABLE X. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 1			
Thermal shock and voltage conditioning 1/	3.10	4.6.6.1 and 4.6.6.2	100% inspection (see table XV)
Voltage conditioning at +85°C 2/	3.10	4.6.6.3	100% inspection (see table XV, PDA overall requirement only)
Radiographic inspection (encapsulated capacitors only)	3.9	4.6.5	100% inspection
Subgroup 2			
Visual and mechanical inspection; material, physical dimensions, design, construction, marking, and workmanship	3.1, 3.4, 3.27, and 3.28	4.6.3	20 samples 0 failures
Subgroup 3			
Destructive physical analysis	3.15	4.6.11	See table XIII
Subgroup 4			
Body insulation (CKS26 only)	3.13	4.6.9.2	20 samples 0 failures

1/ For solder coated nonleaded capacitor chips, thermal shock and voltage conditioning in accordance with 4.6.6.1 and 4.6.6.2 may be performed prior to addition of final metallization; and if performed, shall not be required during group A.

2/ Voltage conditioning at +85°C is only required for solder coated nonleaded capacitor chips that had voltage conditioning performed prior to final metallization.

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4.5.2.3 Group B inspection. Group B inspection shall consist of the tests specified in table XI and shall be performed on sample units from lots which have been subjected to and have passed group A inspection. Copies of group B data shall be forwarded to purchaser with parts. Parts may not be shipped until the conclusion of 1,000 hours of life test.

TABLE XI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Sample size	Accept/ reject
Subgroup 1				
Thermal shock	3.10	4.6.6.1	Table XVII	Table XVII
Life test	3.23	4.6.19	Table XVII	Table XVII
Subgroup 2				
Humidity, steady-state, low voltage	3.20	4.6.16.1	12	0
Subgroup 3				
Voltage-temperature limits (same samples)	3.19	4.6.15	12	1
Moisture resistance (same samples)	3.20	4.6.16.2		
Subgroup 4				
Equivalent series resistance (UHF)	3.24	4.6.20	12	0
Equivalent series resistance (RF)	3.25	4.6.21		
Resonance frequency	3.26	4.6.22		

4.5.3 Periodic inspection. Periodic inspection shall consist of group C inspection. Except where the results of this specification show noncompliance with the applicable requirements, delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections. Group C data shall be submitted to the qualifying activity every 6 months.

4.5.3.1 Group C inspection. Group C inspection shall consist of the tests specified in table XII in the order shown. Samples having group C testing shall be selected from lots that have passed group A and have been submitted for group B inspection. Separate samples of nonleaded, dual-in-line package, and samples of other axial and radial leaded capacitors shall be selected every 2 months in accordance with table XII. Capacitance values manufactured during the period shall be represented as far as practical in at least the approximate ratio of production.

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TABLE XII. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of units to be inspected	Number of defectives permitted
Subgroup 1 - Leaded devices				
Terminal strength	3.16	4.6.12.1	12	1
Solderability	3.17.1	4.6.13.1	12	
Resistance to soldering heat	3.18.1	4.6.14.1		
Resistance to solvents	3.22	4.6.18		
Subgroup 2 - Chip devices				
Terminal strength	3.16	4.6.12.2	12	1
Solderability	3.17.2	4.6.13.2	12	
Resistance to soldering heat	3.18.2	4.6.14.2		

4.5.3.2 Disposition of sample units. Sample units which have been subjected to groups B and C inspections shall not be delivered on the contract. These samples shall be maintained by the manufacturer for a minimum of 15 years.

4.5.3.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection) or the inspection which the original sample failed, at the option of the qualifying activity. Group A (and group B if applicable) inspection may be reinstituted; however, final acceptance shall be withheld until the group C inspection has shown that corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-39028.

4.6 Methods of inspection.

4.6.1 Nondestructive internal examination (see 3.5).

4.6.1.1 Neutron radiography (see 3.5.1). Capacitors shall be subjected to neutron radiography as specified in appendix A and as defined in baseline documentation (see 3.3).

4.6.1.2 Ultrasonic examination (see 3.5.2). Capacitors shall be subjected to ultrasonic examination as specified in appendix A and as defined in baseline documentation (see 3.3).

4.6.2 Pretermination destructive physical analysis (see 3.6). Capacitors shall be examined as specified in EIA standard RS-469 and in accordance with table XIII.

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TABLE XIII. Destructive physical analysis sample size.

Lot size	Minimum sample size	Accept	Reject
1 - 500	14	0	1
501 - 10,000	32	1	2
10,001 - 35,000	50	2	3
35,001 - 500,000	80	3	4

4.6.3 Visual examination (see 3.7). Capacitors shall be 100 percent examined under magnification as specified in the applicable appendix and shall meet the visual requirements of 3.28 for leaded capacitors and appendix C for nonleaded capacitors.

4.6.4 Pre-encapsulation terminal strength (leaded capacitors only) (see 3.8). Capacitors shall be examined as specified in appendix D and in accordance with table XIV.

TABLE XIV. Pre-encapsulation terminal strength.

Lead configuration	Lead pull minimum	Sample size
Radial per lead	1.8 kg (4.0 lbs.)	10 or 2 percent 1/
Axial per lead	1.8 kg (4.0 lbs.) all styles except CKS11 and CKS12 0.9 kg (2.0 lbs.)	10 or 2 percent 1/
Dual-in-line package	2.14 kg (4.7 lbs.)	10 or 2 percent 1/

1/ Whichever is greater, up to 32 pieces maximum.

4.6.5 Radiographic inspection (encapsulated capacitors only). The leaded capacitors shall be radiographed in accordance with 3.9 and appendix E. The following details shall apply:

a. Number of views: (see appendix E, figure E-2).

(1) Radial lead style capacitors: One view perpendicular to the plane of the leads.

(2) Axial lead style capacitors: Two views perpendicular to the plane of the leads, with the second view made after rotation of the capacitor 90° around its own axis.

b. Inspection of films: The films shall be inspected on a back-lighted illuminator using a minimum of 10X magnification.

c. Defects: Capacitors containing any defects as specified in 3.9 and appendix E, figure E-3, shall be rejected.

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d. Summary data of inspection results: A summary of the results of radiographic inspection (number of parts inspected, number accepted, number rejected) shall be included on the report of group A inspection furnished with each shipment of capacitors.

e. Retention of films: Radiograph films and test reports shall be retained by the manufacturer for a minimum period of 15 years.

4.6.6 Thermal shock and voltage conditioning (see 3.10). Capacitors shall be subjected to the tests of 4.6.6.1 and 4.6.6.2, as applicable (see tables VIII, X, and XI).

4.6.6.1 Thermal shock. Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition A, except that in step 3, sample units shall be tested at +125°C.

b. For qualification testing: 100 cycles of thermal shock.

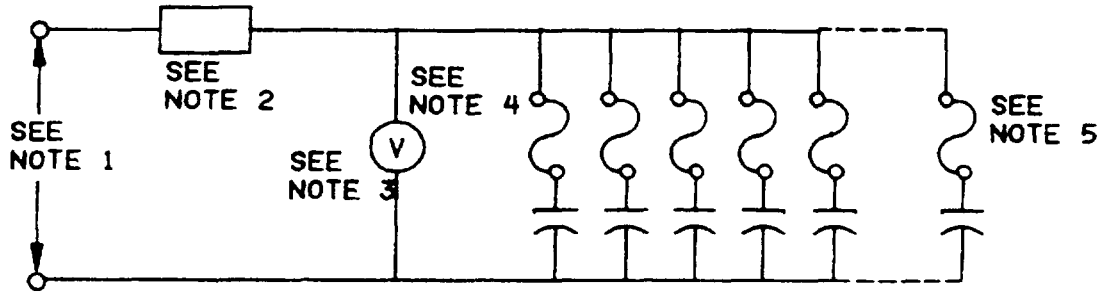
c. For group A testing: 20 cycles of thermal shock.

d. For group B testing: 100 cycles of thermal shock.

4.6.6.2 Voltage conditioning (see 3.10). One of the voltage conditioning tests in 4.6.6.2.1 or 4.6.6.2.2 shall be performed. The lot traveler shall indicate which test is used. When the optional voltage conditioning test of 4.6.6.2.2 is used, the traveler shall include the specific accelerated voltage used and the test time.

4.6.6.2.1 Standard voltage conditioning (see 3.10). Standard voltage conditioning shall be started after completion of the thermal shock test. The voltage conditioning shall consist of applying twice the rated voltage to the units at the maximum rated temperature of +125°C for a minimum of 168 hours and a maximum of 264 hours. The voltage conditioning may be terminated at any time during the 168-hour to 264-hour time interval that failures (blown fuses or less than 95 percent voltage) meet the requirements for the PDA during the last 48 hours listed in table XV. Voltage shall be applied and shall reach maximum value within 1 second, maximum. To assure that at least 95 percent of the applied test voltage is maintained for the duration of the exposure period, the circuit on figure 4 shall be used. After completion of the exposure period, the unit shall be allowed to stabilize at room temperature (+25°C). After stabilization at room temperature, the dielectric withstanding voltage and insulation resistance shall be measured as specified in 4.6.9.1 and 4.6.10, respectively. After measurement of dielectric withstanding voltage and insulation resistance at +25°C, the unit shall be stabilized at maximum rated temperature and insulation resistance measured. After allowing the unit to stabilize at room temperature for 10 hours minimum, the capacitance and dissipation factor shall be measured as specified in 4.6.7 and 4.6.8, respectively. If the voltage conditioning test is performed with individual fuses in series with each part, any part tested in a position where a fuse fails shall be tested for insulation resistance and dielectric withstanding voltage. If the part meets the initial requirements for insulation resistance and dielectric withstanding voltage, the part shall be rejected but shall not count against the percent defective allowed in table XV.

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NOTES:
1. The power supply shall be capable of providing a minimum of twice the rated voltage of the capacitors at a fault current of 30 mA minimum.
2. The current limiting device shall be a resistor and/or a fuse. The current shall be limited to no less than 30 mA and no more than 10 A.
3. There shall be a voltage monitor that will trigger an alarm and shut off the test if the applied voltage drops by more than 5 percent.
4. Fuses shall be optional and resistors shall not be allowed. If fuses are used, the total resistance of the fuse, connections, and the connecting wires shall not exceed 50 ohms at +125°C.
5. The capacitor bank shall be no less than 10.

FIGURE 4. Voltage conditioning circuitry.

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TABLE XV. Percent defectives allowable for subgroup 1, group A inspection.

Style	PDA last 48 hours during voltage conditioning at +125°C	PDA overall including voltage conditioning at +85°C
CKS05	1 unit or 0.1 percent	3 percent
CKS05S	1 unit or 0.1 percent	3 percent
CKS06	1 unit or 0.2 percent	5 percent
CKS06S	1 unit or 0.2 percent	5 percent
CKS07	1 unit or 0.2 percent	5 percent
CKS26	1 unit or 0.2 percent	5 percent
CKS11	1 unit or 0.1 percent	3 percent
CKS12	1 unit or 0.1 percent	3 percent
CKS14	1 unit or 0.1 percent	3 percent
CKS15	1 unit or 0.2 percent	5 percent
CKS16	1 unit or 0.2 percent	5 percent
CKS51	1 unit or 0.1 percent	3 percent
CKS52	1 unit or 0.1 percent	3 percent
CKS53	1 unit or 0.1 percent	3 percent
CKS54	1 unit or 0.1 percent	3 percent
CKS28	1 unit or 0.1 percent	3 percent
CKS29	1 unit or 0.1 percent	3 percent
CKS22	1 unit or 0.1 percent	3 percent
CKS23	1 unit or 0.1 percent	3 percent
CKS24	1 unit or 0.1 percent	3 percent

4.6.6.2.2 Optional voltage conditioning. The manufacturer, with approval from the qualifying activity, may perform an optional voltage conditioning test instead of the standard voltage conditioning test of 4.6.6.2.1. All conditions of 4.6.6.2.1 apply, with the exception of the voltage applied and the test time. The minimum time duration, T(test), shall be calculated as follows:

$$T\,(test) = \frac{800}{(E\ test/E\ rated)^3}$$

Where: $2 \times E\ rated \leq E\ test \leq 4 \times E\ rated$

T(test) = Minimum test time in hours

E test = Applied voltage

E rated = Rated voltage of the capacitor

4.6.6.3 Voltage conditioning at +85°C for nonleaded solder coated chip capacitors (see 3.10). Voltage conditioning (see table X) shall consist of applying twice the rated voltage to the units at +85°C for 48 hours. The voltage conditioning circuit (see figure 4) and measurements of capacitance, dissipation factor, dielectric withstanding voltage, and insulation resistance shall be the same as that described in 4.6.6.2.

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4.6.7 Capacitance (see 3.11). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following detail and exception shall apply:

a. Test frequency: 1 megahertz ±100 kHz when the nominal capacitance is 100 pF or less for BX and BR characteristics, and 1,000 pF or less for BP and BG characteristics. One kHz ±100 Hz when the nominal capacitance is greater than above.

b. Voltage: A root-mean-square potential of 1.0 ±0.2 volt.

NOTE: Following a dielectric withstanding voltage or insulation resistance test, capacitance measurement may be delayed for a period of up to 24 hours.

4.6.8 Dissipation factor (see 3.12). The dissipation factor shall be measured with a capacitance bridge or other suitable method at the frequency and voltage specified in 4.6.7. The inherent accuracy of the measurement shall be ±2 percent of the reading plus 0.1 percent dissipation factor (absolute) unless otherwise specified. Suitable measurement techniques shall be used to minimize errors due to the connections between the measuring apparatus and the capacitor.

4.6.9 Dielectric withstanding voltage (see 3.13).

4.6.9.1 Dielectric. Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Magnitude and nature of test voltage: 250 percent of the direct current rated voltage.

b. Duration of application of test voltage: 5 ±1 seconds. The test voltage shall be raised from 0 to the specified value within 1 second, maximum.

c. Points of application of test voltage: Between the capacitor-element terminals.

d. Limiting value of surge current: Shall be limited between 30 and 50 milliamperes.

e. Examination after test: Capacitors shall be examined for evidence of damage and breakdown.

4.6.9.2 Body insulation. Leaded capacitors shall be tested as specified in 4.6.9.1 with the following exception: Points of application of test voltage: The encapsulated body of the capacitors shall be brought into intimate contact with a conductive material not less than .0625 inch (1.588 mm) and not more than .125 inch (3.18 mm) away from the lead wires. A direct current potential of 500 V dc shall be applied between the two leads connected together and the tape or foil for a period of 5 ±1 seconds. The test circuit shall be so arranged that the surge current does not exceed 50 mA.

4.6.10 Insulation resistance (see 3.14). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

a. Test potential: Rated voltage.

b. Special conditions: If a failure occurs at a relative humidity above 50 percent, the insulation resistance may be measured again at any relative humidity less than 50 percent.

c. Points of measurement: Between the mutually insulated points.

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4.6.11 Destructive physical analysis (see 3.4.1 and 3.15). Capacitors shall be examined as specified in EIA Standard RS-469 and in accordance with table XIII.

4.6.12 Terminal strength (see 3.16).

4.6.12.1 Terminal strength (leaded capacitors) (see 3.16). Capacitors shall be tested in accordance with method 211, MIL-STD-202. The following details and exceptions shall apply:

a. For dual-in-line package (DIP) capacitors, bend leads to full radial (down) position before testing.

b. Test condition A:

(1) Method of holding: Capacitors shall be held by one terminal and the load shall be applied gradually to the other terminal.

(2) Applied force: 2 ±.1 kilograms.

c. Test condition C: Applicable to radial-lead and DIP units only.

d. Test condition D: Applicable to axial-lead units only.

e. Examination after test: Capacitors shall be visually examined for evidence of loosening or rupturing of the terminals.

4.6.12.2 Terminal strength (nonleaded capacitors) (see 3.16). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

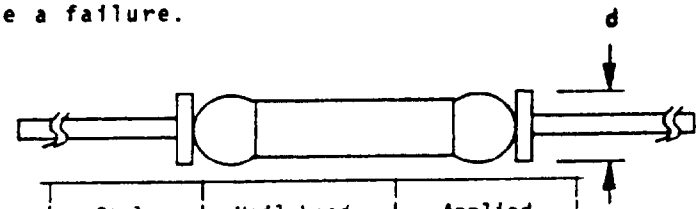
a. Test condition A:

(1) Capacitors shall have nail head, lead wire (.020 inch to .025 inch) (.51 mm to .64 mm) attached as shown on figure 5, prior to pull test.

(2) Method of holding: Capacitors shall be held by one terminal and the load shall be applied gradually to the other terminal.

(3) Applied force: See figure 5.

b. Examination after test: Capacitors shall be visually examined for evidence of loosening or rupturing of the terminals. Breakage of lead wire or solder interface, which does not expose ceramic, shall not constitute a failure.



Style	Nail head diameter d	Applied force
CKS51	.040 ±.005	1 Kg ±.1
CKS52	.040 ±.005	1 Kg ±.1
CKS53	.045 ±.005	2 Kg ±.2
CKS54	.045 ±.005	2 Kg ±.2

Inches	mm
.005	0.13
.040	1.02
.045	1.14

FIGURE 5. Lead attachment for chip pull test.

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4.6.12.3 Pre-encapsulation terminal strength (dual-in-line package (DIP) capacitors) (see 3.16). Capacitors shall be tested in accordance with appendix D.

4.6.13 Solderability (see 3.17).

4.6.13.1 Solderability for leaded capacitors (see 3.17.1). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. Two terminations shall be tested.

4.6.13.2 Solderability for nonleaded capacitors (see 3.17.2). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details and exceptions shall apply:

a. Solder shall be Sn62 in accordance with QQ-S-571.

b. Each chip capacitor shall be immersed in molten solder to a depth of .020 +.010, -.000 inch.

c. Number of terminations to be tested: Two.

d. Examination of terminations shall be in accordance with 3.17.2. In case of dispute, the percent coverage with pinholes or exposed termination material shall be determined by actual measurement in these areas, as compared to the total area.

4.6.14 Resistance to soldering heat (see 3.18).

4.6.14.1 Resistance to soldering heat for leaded capacitors (see 3.18.1). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Capacitors shall be immersed using procedure 1.

b. Depth of immersion in the molten solder: To within 0.050 inch from the capacitor body (see figure 6).

c. Test condition B: Except that the immersion duration shall be 20 ±1 seconds.

d. Cooling time prior to measurement after test: 10 +1, -0 minutes.

e. Measurements before test: Capacitance, DF and IR at +25°C shall be measured and recorded as specified in 4.6.7, 4.6.8, and 4.6.10, respectively.

f. Measurements after test: Capacitors shall meet the requirements of 3.18.1. Capacitance, dissipation factor, and insulation resistance at +25°C shall be measured and recorded as specified in 4.6.7, 4.6.8, and 4.6.10, respectively.

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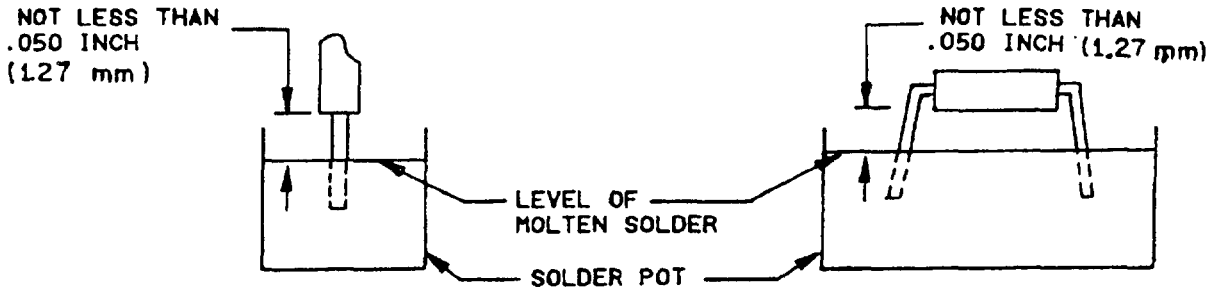


FIGURE 6. Depth of immersion in molten solder.

4.6.14.2 Resistance to soldering heat (nonleaded capacitors) (see 3.18.2). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

a. Capacitors shall be immersed using procedure 1, two times.

b. Test condition B: Except the solder temperature shall be +230°C, +5°C, -0°C, and the duration shall be 5, +1, -0 seconds.

c. Cooling time prior to measurement after test: 10 minutes minimum and up to 24 hours maximum.

d. Measurements before test: Capacitance, dissipation factor, and insulation resistance at +25°C shall be measured and recorded as specified in 4.6.7, 4.6.8, and 4.6.10, respectively.

e. Examination and measurement after test: Capacitors shall meet the requirements of 3.18.2. Capacitance, dissipation factor, and insulation resistance at +25°C shall be measured and recorded as specified in 4.6.7, 4.6.8, and 4.6.10, respectively.

4.6.15 Voltage-temperature limits (see 3.19). The temperature of each capacitor shall be varied as specified in table XVI. Capacitance measurements shall be made at the frequency and voltage specified in 4.6.7. The direct current rated voltage specified in table XVI shall be maintained on the capacitor during steps E to G inclusive. Capacitance measurements shall be made at each step specified in table XVI and at five intermediate temperatures between steps B and G to establish a true characteristic curve. Capacitors shall be kept at each temperature until temperature equilibrium is attained.

4.6.15.1 Voltage-temperature limits for quality conformance inspection. Capacitance measurements shall be made as specified in 4.6.7, except that the measurements shall be made only for steps C, D, E, and G of table XVI.

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TABLE XVI. Voltage-temperature limit cycle.

Step	Voltage, dc	Temperature °C
A	None	+25 ±2
B	None	-55 ±2
C 1/	None	+25 ±2
D	None	+125 ±2
E	Rated	+125 ±2
F	Rated	+25 ±2
G	Rated	-55 ±2

1/ Reference point.

4.6.16 Moisture resistance (see 3.20).

4.6.16.1 Humidity, steady state, low voltage. Capacitors shall be tested in accordance with method 103, condition A of MIL-STD-202. The following details and exceptions shall apply. Note: At no time during test shall voltage greater than 1.5 volts be applied to any capacitor under test.

a. Initial measurements: Capacitance.

b. Test: Capacitors shall be subjected to an environment of +85°C with 85 percent relative humidity for 240 hours minimum. Cycling shall not be performed. A dc potential of 1.3 ±0.25 volts shall be applied continuously through a 100,000-ohm resistance to each device under test.

c. Final measurements: Upon completion of above test, remove capacitors from chamber and allow 3 hours, 30 minutes, ±30 minutes to dry and stabilize at +25°C before performing insulation resistance (IR), through a 100 kilohm resistor at 1.3 ±0.25 volts, capacitance and dissipation factor in accordance with 4.6.10, 4.6.7, and 4.6.8, respectively.

d. Visual examination: After final measurements, the capacitors shall be examined for evidence of mechanical damage and obliteration of marking.

e. Leads may be attached to chip capacitors for mounting and loading purposes. Mechanical loading is acceptable.

4.6.16.2 Moisture resistance, normal voltage. Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following detail and exceptions shall apply:

a. Initial measurements: Capacitance.

b. Number of cycles: 20 continuous cycles.

c. Tests: During the first 10 cycles only, a direct current potential of 50 volts shall be applied across the capacitor terminals. Once each day, a check shall be performed to determine whether a capacitor has shorted. Vibration cycle of step 7b shall not be performed.

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d. Final measurements and examination: On completion of step 6 of the final cycle, capacitors shall be measured for capacitance, dielectric withstanding voltage, and insulation resistance as specified in 4.6.7, 4.6.9, and 4.6.10, respectively.

e. Leads may be attached to chip capacitors for mounting and loading purposes. Mechanical holding is acceptable.

4.6.17 Vibration, high frequency (leaded capacitors only) (see 3.21). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Capacitors shall be rigidly mounted on a mounting fixture by the body. The leads shall be secured to rigidly supported terminals, so spaced that the length of each lead from the capacitor is approximately 0.375 inch (9.52 mm) when measured from the edge of the supporting terminals. Leads shall be within 15° of being parallel. When securing leads, care shall be taken to avoid pinching the leads. The mounting fixture shall be so constructed as to preclude any resonances within the test range. An examination of the mounting fixture shall be made on a vibrator. If any resonant frequencies are observed, adequate steps must be taken to damp the structure.

b. Electrical load conditions: During the test, a direct current potential equal to 125 percent of the direct current rated voltage (see 3.1) shall be applied between the terminals of the capacitor element under test.

c. Test condition E (50's): Except frequency range of 10 through 3,000 Hz.

d. Duration and direction of motion: Equal amounts of time in each of three mutually perpendicular planes (total of 8 hours).

e. Measurements during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration, or open- or short-circuiting.

f. Examination after vibration: Capacitors shall be visually examined for evidence of mechanical damage.

4.6.18 Resistance to solvents (see 3.22). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

a. The marked portion of the capacitor body shall be brushed.

b. The number of sample units to be examined shall be as specified in table VIII.

c. Capacitors shall be visually examined for evidence of mechanical damage and obliteration of marking.

4.6.19 Life (at elevated ambient temperature) (see 3.23). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Minimum sample size and accept/reject criteria shall be in accordance with table XVII.

b. Distance of temperature measurements from specimens, in inches, not applicable.

c. Test temperature and tolerance: +125°C, +4°C, -0°C.

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d. Capacitors shall be subjected to the voltage and circuit specified in 4.6.6.2.

e. Test duration: 4,000 hours for qualification and 1,000 hours for group B.

f. Measurements during and after exposure: At the conclusion of 250, 1,000, 2,000, and 4,000 hours, and while the capacitors are at the applicable high-test temperature, the insulation resistance shall be measured as specified in 4.6.10. At the option of the manufacturer, the units may be immediately transferred (period of transfer not to exceed 15 minutes) to another chamber maintained at the same temperature for the purpose of measuring insulation resistance. The insulation resistance measurement shall be made only after the units have stabilized at the test temperature. The capacitors shall then be returned to the inspection conditions specified in 4.3 and shall be visually examined for evidence of mechanical damage and obliteration of marking; capacitance, dissipation factor, and insulation resistance shall be measured as specified in 4.6.7, 4.6.8, and 4.6.10, respectively. Equivalent series resistance shall be measured for high frequency styles only, as specified in 4.6.20 and 4.6.21.

TABLE XVII. Life test.

Lot size	Minimum sample size	250-4,000 hours	
		Accept	Reject
1 - 1,200	25	0	1
1,201 - 10,000	80	1	2
10,001 - 150,000	125	2	3

4.6.20 Equivalent series resistance (UHF) (when specified, see 3.1)(see 3.24). The UHF equivalent series resistance shall be measured in accordance with appendix F, using the wavelength and frequency range specified in table XVIII.

TABLE XVIII. Equivalent series resistance (UHF) test frequency range.

Capacitance range	Wavelength	Frequency range
1-29 pF 1/	7/4	910-1050 MHz
30-100 pF	5/4	640-660 MHz
101-1000 pF	3/4	380-400 MHz

1/ Not practically measurable for capacitance values below 1 pF.

4.6.21 Equivalent series resistance (RF) (when specified, see 3.1), (see 3.25). The RF equivalent series resistance shall be measured in accordance with appendix F at one quarter wavelength in a frequency range of 130 MHz to 260 MHz.

4.6.22 Resonance frequency (when specified, see 3.1) (see 3.26). The resonance frequency shall be measured in accordance with appendix F for capacitance values of 1 pF and above.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-39028.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These capacitors are primarily designed for use in circuits and systems requiring high reliability electronic components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Title, number, and date of the applicable specification sheet, and the complete part number (see 3.1).

d. Lead length, if different from that specified (see 3.1). Lead lengths may be a minimum of 1 inch (25.4 mm) long for use in tape and reel automatic insertion equipment, when specified.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contracts, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Goddard Space Flight Center, National Aeronautics and Space Administration, Parts Project Office (NPPO), Greenbelt, MD 20771; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC -E), Dayton, Ohio 45444. Application for qualification tests shall be made in accordance with SD-6 "Provisions Governing Qualification" (see 6.3.1).

6.3.1 SD-6 "Provisions Governing Qualification". Copies of SD-6 "Provisions Governing Qualification" may be obtained on application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Case insulation. It is not intended that the case insulation be subjected to sustained voltage in excess of 150 percent of the dc rated voltage of the capacitor. Supplementary insulation should be provided where the case may come in contact with higher voltage.

6.5 Subject term (key word) listing.

Ceramic
High frequency
High reliability
Space
Temperature stable

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX A

NONDESTRUCTIVE PHYSICAL ANALYSIS

10. SCOPE

10.1 Scope. This appendix specifies the procedures for the nondestructive internal examination of CKS ceramic capacitors. The test procedures for both neutron radiographic inspection (NRI) and ultrasonic examination are included. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Other publications. The following document forms a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-545 - Determining Image Quality in Thermal Neutron Radiographic Testing.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187)

30. NEUTRON RADIOGRAPHIC INSPECTION (NRI)

30.1 Purpose. This section specifies procedures and practices necessary for the successful NRI of ceramic capacitors for delaminations.

Note: Neutron radiographic inspection of ceramic capacitors for small delaminations (in the order of .001 inch (0.03 mm) or less) require special fixturing and techniques to assure state-of-the-art detection capacity. The prime requirement for this inspection is careful orientation of the plane of delamination parallel to the axis of the neutron beam.

30.2 Requirement. If more than one lot of capacitors is tested in any one fixture, calibrated samples shall be required for each lot.

30.3 Apparatus.

a. A neutron radiographic facility that shall produce a beam comprised basically of unscattered, thermalized neutrons. Gamma radiation, scattered neutrons, and fast neutrons shall be minimized. As a minimum, the beam composition shall produce a radiograph with a ASTM E-545 index of N75-15-11. Maximum gamma content will be 2 percent. The neutron beam shall have an L/D of at least 250.

b. Facility fixturing shall provide for alignment of the capacitor fixture relative to the beam within 0.4°.

c. Radiographic film shall be Eastman type SR-54 or equivalent.

d. Copy film shall be Eastman RP/D or equivalent.

e. Film holder shall be a vacuum cassette with a gadolinium metal converter. Cleanliness is essential to high-quality neutron radiography. The radiography cassettes shall be cleaned prior to each production run. The film shall be loaded into the cassette in a laminar-flow-ultra-clean bench or equivalent.

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f. Capacitor mounting fixture shall be rugged enough to withstand air shipment while securely holding the capacitors in proper orientation. Drawings for suitable fixtures are shown in figures A-1 and A-2. The fixture shall be made of 6061-T6 aluminum and shall be as thin and uniform in thickness as practical in the direction of the neutron beam. Also included in this fixture are aluminum keys and rubber spacer strips. The aluminum keys shall fit tightly in the fixture to assure correct alignment of the capacitors. The rubber spacer strips have .094 inch (2.39 mm), .125 inch (3.18 mm), .188 inch (4.78 mm), and .250 inch (6.35 mm) square cross sections and are used to provide a snug fit of various sized capacitors in the fixture. This fixture must be maintained clean with no tape or gum residue left in the area of interest prior to loading the capacitors.

g. Beam purity indicators and sensitivity gauges shall be in accordance with ASTM E-545. In addition, a defective part standard may be included as an indicator.

h. Radiographic viewer capable of resolving .001 inch (0.03 mm) shall be provided.

i. Two capacitors previously determined to be bad, of the same dielectric material and size of the parts under test, shall be added to the lot for testing.

j. Neutron sensitive labeling shall be used for identification of the capacitors in the fixturing. This material, which is imprintable with a typewriter, is available from the neutron radiographic facility.

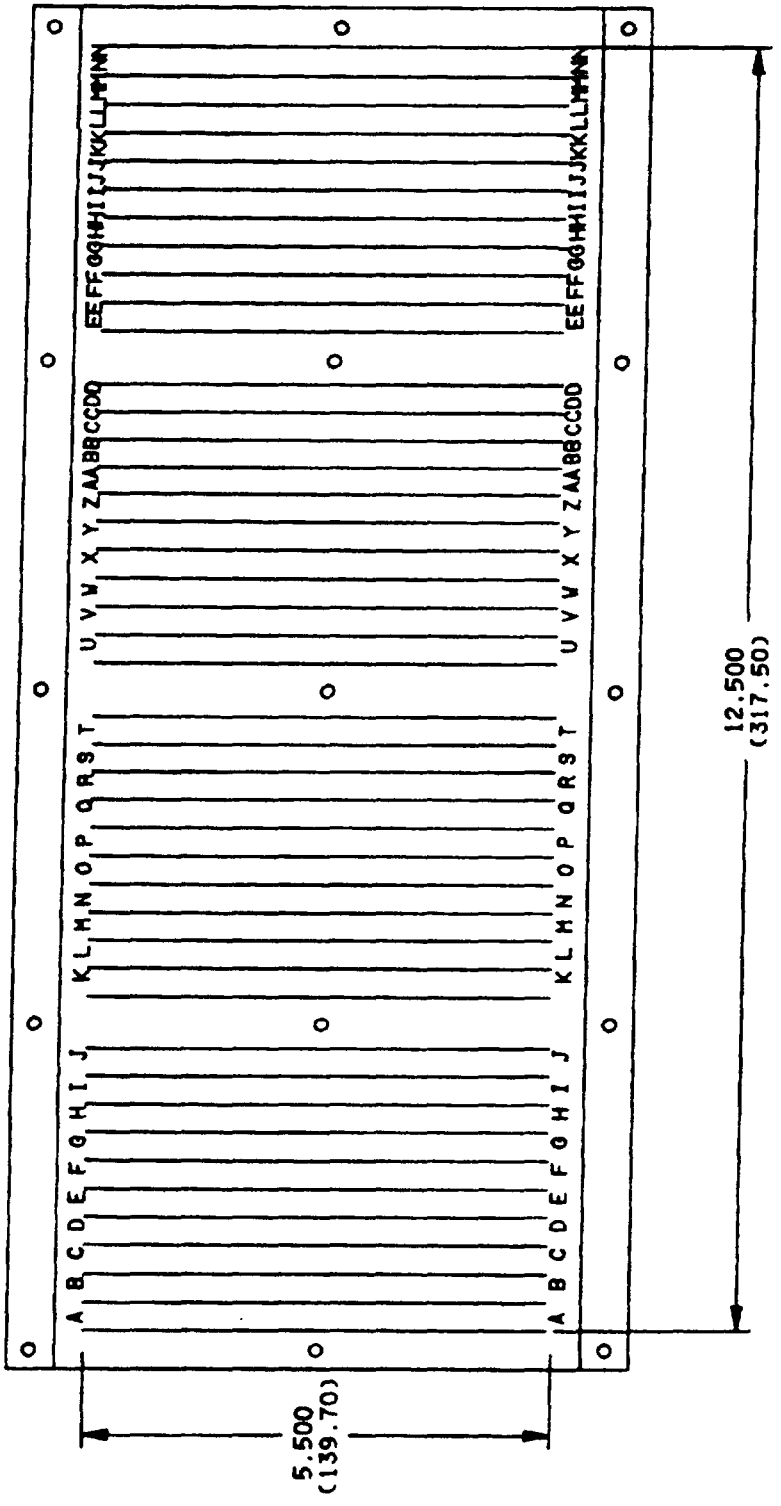
30.4 Procedure. The neutron exposure time shall be adjusted to achieve a film background optical density between 2.0 and 3.0. The lower density shall be used on low neutron attenuation capacitors, the higher density on high attenuation capacitors.

30.4.1 Mounting and views. The capacitors shall be mounted such that the electrodes are parallel both to the neutron beam and to the long axis of the fixture. Alignment keys are required to assure that capacitors remain in a vertical configuration. The alignment keys also provide a visual aid to the film interpreter by providing reference points for counting capacitors. Additionally, rubber spacers may be required to fill gaps and assure that capacitors are held tightly against the alignment keys.

30.4.2 View 3, critical alignment. Each capacitor shall be placed such that the electrodes of the capacitor shall be parallel (±0.6°) to the neutron beam and their longitudinal axes normally shall be perpendicular to the neutron beams. An aluminum key spacer placed every 10 capacitors provides a realignment surface in the event of uneven coatings on capacitors. A row of capacitors shall be not more than 4 inches (102 mm) in length to minimize the effect of the angle of divergence of the neutron beam.

30.4.3 Radiographic quality standard. ASTM beam purity indicators (BPI's) and sensitivity gauges shall be radiographed in accordance with current recommendations or ASTM E-545. In addition, standard defect parts shall also be placed on the plate to be radiographed. These devices shall be used to assure a quality neutron radiograph. In those cases where no room is allowed for the BPI or sensitivity gauge, the performing facility shall certify that the required quality level can be met with the exposure conditions used. If required, a verification blank radiograph shall be supplied, using identical exposure conditions with the BPI and sensitivity gauge.

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NOTE: Dimensions are applicable to the active portion (subject to exposure) only.

FIGURE A-1. Typical N-ray fixture.

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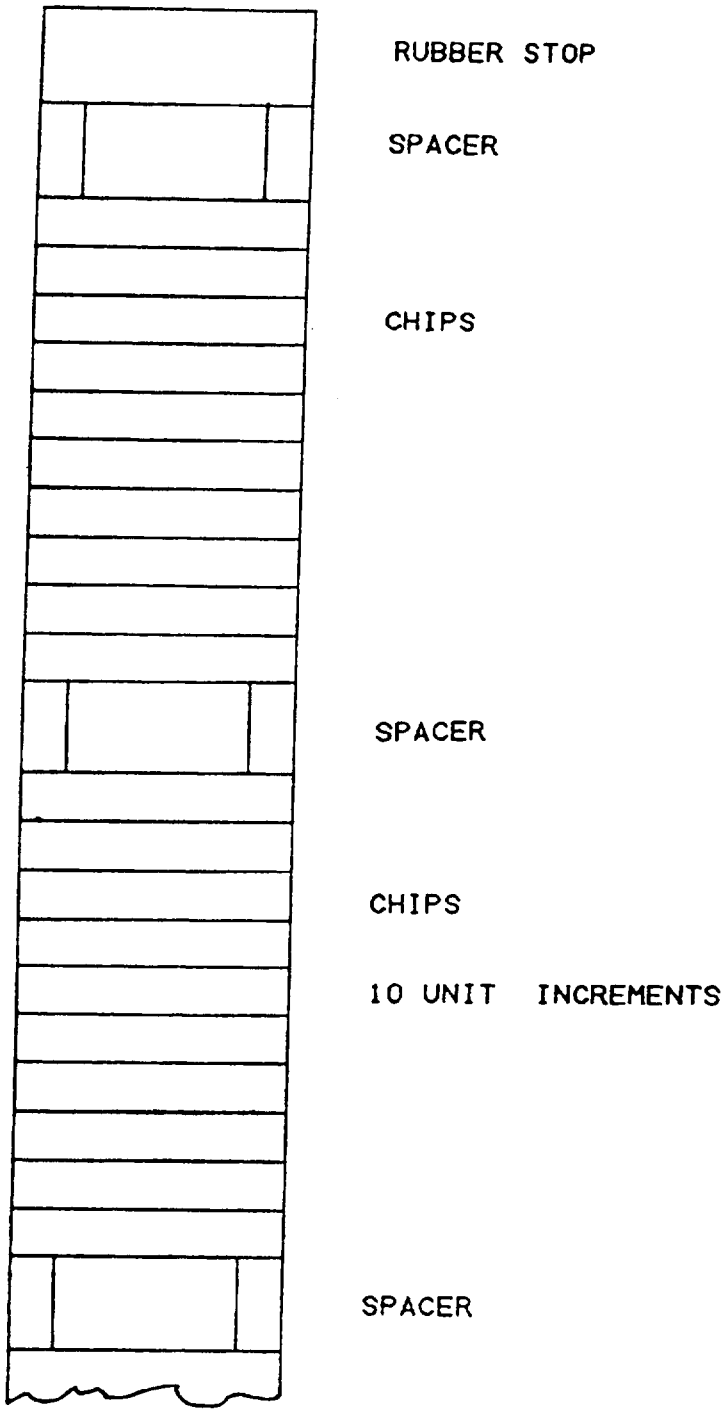


FIGURE A-2. Loading tunnel.

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30.4.4 Film. Each film shall normally bear the following minimum identification typed on neutron absorbing material.

a. Name of performing facility.

b. Date.

c. Film number.

d. Organization for whom test is being performed.

e. Part titles and serialization.

f. User mounted and/or labeled trays will be so noted on the radiograph.

30.4.5 Processing. The radiographic film manufacturer's recommended procedure shall be used to develop the exposed film, and the film shall be processed so that it is free of processing defects such as fingerprints, scratches, fogging, chemical spots, blemishes, etc. Furthermore, no apparent film flaws shall be present in a critical area of the components. Should one occur, the radiograph shall be redone to prove that the flaw is in the film and not in the part. Also, since gadolinium conversion screens with zero surface defects are unavailable, a limited number of bright specks due to these flaws are allowable if positively identified and correlated to a known flaw in the screen.

30.5 Operating personnel. Personnel who will perform radiographic inspection shall have training in radiographic procedures and techniques so that defects revealed by this method can be validly interpreted and compared with applicable standards. The following minimum vision requirements shall apply for visual acuity of personnel inspecting film:

a. Distant vision shall equal at least 20/30 in both eyes, corrected or uncorrected.

b. Near vision shall be such that the operator can read Jaeger type no. 2 at a distance of 16 inches (406 mm), corrected or uncorrected.

30.6 Inspection of radiographs. Utilizing the equipment specified herein, radiographs shall be inspected to determine that each device conforms to this standard and defective devices shall be rejected. Interpretation of the radiograph shall be made under low ambient light level conditions without glare on the radiographic viewing surface. The radiographs shall be examined on a suitable illuminator with variable intensity or on a viewer suitable for radiographic inspection on projection type viewing equipment. The radiograph shall be viewed at a magnification between 6X and 25X. Viewing masks may be used when necessary. Any radiograph not clearly illustrating the features in the radiographic quality standards is not acceptable and another radiograph of the devices shall be taken.

30.7 Reports and records. Appropriate information (number tested, number failed) from scanning results shall be recorded on the chip lot traveler. The scanner printout sheets shall be identified with the chip lot number, size, voltage rating, and capacitance value, and placed on file.

40. ULTRASONIC EXAMINATION TEST METHODS AND REQUIREMENTS

40.1 Equipment. The test equipment shall be in accordance with the list below or its equivalent:

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a. Ultrasonic scanner US450 series 1/.

b. Pulser/receiver unit model 50S 1/.

c. Lab scanner model 450 1/.

d. 25 MHz Transducer model 57 AH006 1/.

e. Sectioning equipment EIA Standard RS-469.

f. Glass plate.

g. Plexiglass plate.

h. Capacitor chips known to be good (absence of delamination and cracks).

i. Capacitor chips known to be defective (with delaminations).

40.2 Requirements. Capacitor chips determined to be good after testing and when sectioned in accordance with paragraph 4.6.2 must meet the DPA requirements of paragraph 4.6.2.

40.2.1 Lots tested. Only one production lot of parts shall be scanned at one time.

40.2.2 Calibration samples. Two previously determined bad capacitors of the same dielectric material and size of the parts under test shall be added to the lot for testing.

40.3 Initial setup procedure. The initial setup procedure is as follows:

a. Equipment shall be turned on and warmed up for 15 minutes, minimum.

b. The tank shall have a water depth of 6 ±.250 inches (152 ±6.35 mm). 175 ±5 ML of wetting agent shall be added to the tank.

c. Chips shall be placed in identifiable rows on stiff plastic or glass trays. The tray must be level and perpendicular to the transducer head axis. The tray shall be checked after it is in the tank to assure that no air bubbles are trapped under the chips, or tray, or between the chips.

40.4 Operating procedures. The chip image return, and gate alarm setting shall be adjusted each time the equipment is turned off and on the first tray of chips to be scanned per lot. Adjustments shall be in accordance with the test facility's operating instructions. The adjustments shall clearly confirm the good and bad calibration samples. These parts shall be easily identifiable from the remainder of the lot under test.

40.5 Qualification. Each test facility shall have a responsible person who has successfully completed a nondestructive testing course given by the manufacturers of the ultrasonic test equipment. The ultrasonic test equipment shall not be operated by anyone who has not been certified and instructed in the use of ultrasonic testing by the designated person. The responsible person shall prepare a test method that describes the general procedures, requirements, and techniques for ultrasonic scanning of ceramic chips.

1/ Manufactured by Automation Industries, Inc., Sperry Products Division, Subsidiary of General Cable.

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40.6 Equipment usage. Testing shall be done in accordance with the test equipment manufacturer's instructions and the test facility's operating procedures. The tank must be completely drained, cleaned, and refilled at least once a month. NOTE: Anytime the tank is drained and refilled, the transducer lens must be checked to ensure bubbles have not been trapped.

a. The scanners (XY recorders) shall be activated and the printout of the calibration samples shall be compared against the scanner's "0" scope. If correlation for the calibration samples is obtained with the setting of 40.4, the entire lot can be scanned with that setting.

b. Any questionable scope results shall reject the chip under question.

40.7 Documentation. Appropriate information (number tested, number failed) from scanning results shall be recorded on the chip lot traveler. The scanner printout sheets shall be identified with the chip lot number, size, voltage rating, and capacitance value, and placed on file.

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APPENDIX B

PRE-ENCAPSULATION VISUAL INSPECTION CRITERIA FOR LEADED CAPACITORS

10. SCOPE

10.1 Scope. This appendix specifies the visual inspection criteria for leaded capacitors. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. Applicable documents. This section is not applicable to this appendix.

30. PROCEDURES FOR INSPECTION AND REJECTION

30.1 Method of inspection. Each device shall be examined under 7 power minimum magnification to determine compliance with the requirements specified herein.

30.2 Rejection criteria for both axial and radial leaded devices. Devices that deviate from the material, design, or construction requirements specified, or that fail to meet the following requirements or those specified in 40 shall be unacceptable.

30.2.1 Chip examination.

a. Cracks in the ceramic (thermal shock cracks).

b. Open delaminations in chips which expose internal electrode material.

c. "Cold" solder joints (frosted or granular in appearance).

d. Solder bridging or potential bridging between opposing terminations.

e. Excess solder.

f. No solder balls on leads.

40. SPECIFIC LEAD REJECTION CRITERIA

40.1 Axial leaded devices - criteria/assessment.

a. Unit A: Ideal lead to element union. Acceptable (see figure B-1).

b. Unit B: Solder build-up (fillet) on one side only. Acceptable (see figure B-1).

c. Unit C: No solder build-up (fillet), but good element to chip interface. Acceptable (see figure B-1).

d. Unit D: Edge of lead wire does not overhang the edge of the element, and full solder coverage is evidenced. Acceptable (see figure B-2).

e. Unit E: Gap in solder joint under the nail head which reduces lead-to-element interface to less than 90 percent coverage. Rejectable (see figure B-3).

f. Unit F: Gap in solder joint under the nail head which reduces lead-to-element interface to less than 90 percent coverage. Rejectable (see figure B-3).

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g. Unit G: Gaps or voids in solder joint under the nail head which reduces lead-to-element interface to less than 90 percent coverage. Rejectable (see figure B-3).

h. Unit H: Gap in solder joint under the nail head which reduces lead-to-element interface to less than 90 percent coverage. Rejectable (see figure B-3).

i. Unit I: Diameter of lead wire overhangs side of element. Rejectable (see figure B-3). (Note: Nail head may overhang.)

j. Unit J: Edge of lead wire does not overhang edge of element, but gap in solder joint under the nail head reduces lead-to-element interface to less than 90 percent coverage. Rejectable (see figure B-3).

k. Unit K: Center line of nail head tilted greater than 30° beyond center line of chip. Rejectable (see figure B-3).

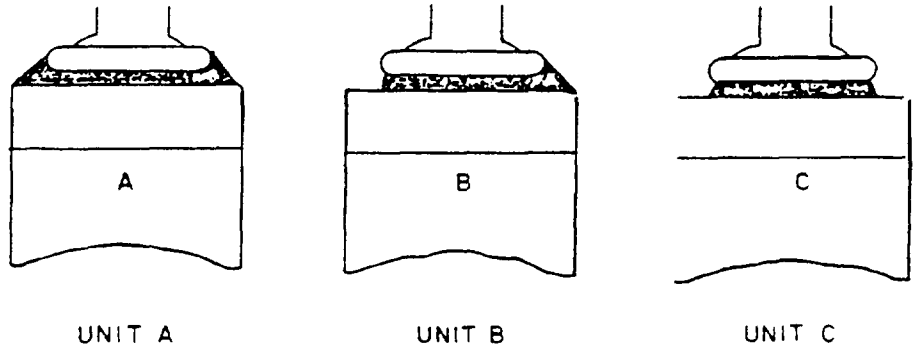


FIGURE B-1. Acceptable units.

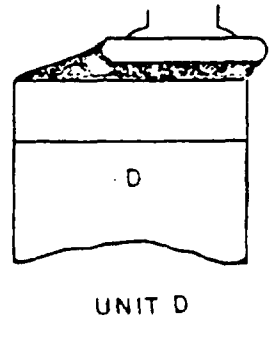


FIGURE B-2. Acceptable only by QC department.

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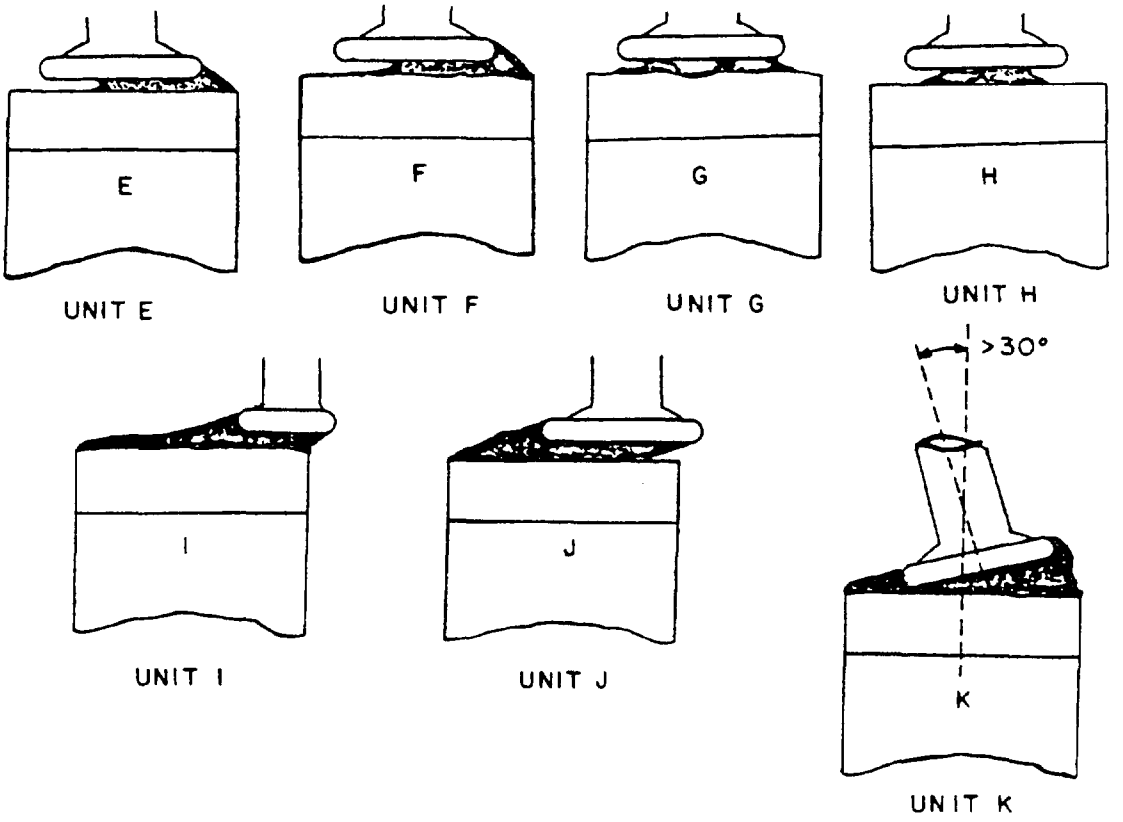


FIGURE B-3. Rejectable devices.

40.2 Radial leaded devices (molded) - criteria/assessment.

a. Unit M: Ideal condition for a radial leaded device prior to encapsulation. Acceptable (see figure B-4).

b. Unit N: Element thickness is greater than or equal to the average lead wire flat dimension. The element must be positioned such that the lead wire flat does not extend more than one-third beyond the thickness of the element. This example is rejectable (see figure B-5).

c. Unit O: Element thickness is less than the average lead wire flat dimension. The element must be positioned such that the lead to element interface is a minimum of 80 percent of the available contact area. This example is rejectable (see figure B-5).

d. Unit P: Angle between chip centerline and lead centerline shall not exceed 30° (see figure B-5).

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e. Unit Q: Ideal chip alignment (top view). Acceptable (see figure B-6).

f. Unit R: If the element thickness is greater than or equal to the average lead wire flat dimension, the element must be positioned such that the lead wire flat does not extend more than one-third beyond the thickness of the element. If the element thickness is less than the average lead wire flat dimension, the element must be positioned such that the lead to element interface is a minimum of 90 percent of the available contact area. This example is rejectable (top view) (see figure B-6).

g. Unit S: This example shows various rejectable characteristics which may be observed on the radial leaded device (see figure B-7). Description of the numbered defects are as follows:

(1) Lead-to-element interface of less than 80 percent is not acceptable.

(2) Cut, dent, or smash in the lead wire exceeding .050 inch (1.27 mm) in length or .005 inch (0.13 mm) in depth is rejectable.

(3) Element is not properly seated in the "saddle". Gaps in excess of .015 inch (0.38 mm) are rejectable.

(4) Chip-outs in the ceramic surface which expose electrodes are not acceptable.

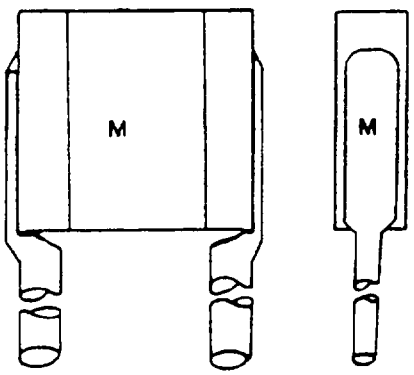


FIGURE B-4. Ideal condition (acceptable unit).

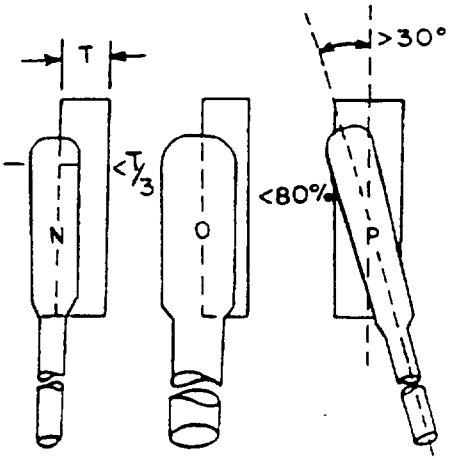


FIGURE B-5. Rejectable units.



FIGURE B-6. Chip alignment.

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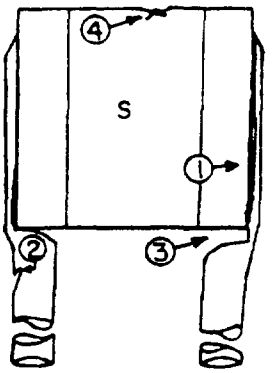


FIGURE B-7. Rejectable characteristics.

40.3 Radial leaded devices (epoxy coated) - criteria/assessment (see figure B-8).

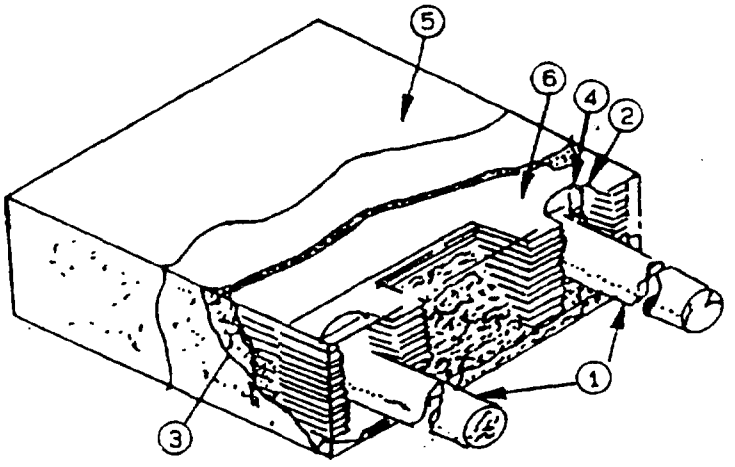
a. Leads shall be 100 percent covered with solder.

b. There shall be no voids under the lead head.

c. The solder fillet shall not extend more than one lead diameter in distance down the lead from the bottom edge of the chip.

d. There shall be no crack or separation between the lead and chip.

e. Leads shall not be off center by more than .015 inch (0.38 mm).



1. Lead wire
2. Solder
3. Dielectric
4. End silver frit
5. Case
6. Electrode

FIGURE B-8. Typical (nonmolded) device with epoxy coat.

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APPENDIX C

VISUAL INSPECTION CRITERIA FOR CHIP CAPACITORS

10. SCOPE

10.1 Scope. This appendix specifies the visual inspection criteria for chip capacitors. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. PROCEDURES FOR INSPECTION AND REJECTION

30.1 Method of inspection. Each device shall be examined under 10 power magnification to determine compliance with the requirements specified herein.

30.2 Rejection criteria. Devices that deviate from the material, design or construction requirements specified, or that fail to meet the following requirements, shall be unacceptable.

30.2.1 Termination metallization.

a. End terminals shall be completely covered. Pinholes less than or equal to .005 inch (0.13 mm) in diameter are permitted (maximum of three pinholes in each surface area).

b. Gaps in the metallization band shall only be acceptable if the metallization band is not less than the minimum metallization band requirement specified (see figure C-1).

c. Metallized edges shall not be reduced to less than 90 percent due to chipping or metallizing process (see figure C-2).

d. There shall be no excess metallization or solder tear which violates the minimum pad separation (see figure C-3).

e. There shall be no foreign material visible adhering to the solder, or voids in the solder revealing greater than 10 percent of the base metallization (see figure C-4).

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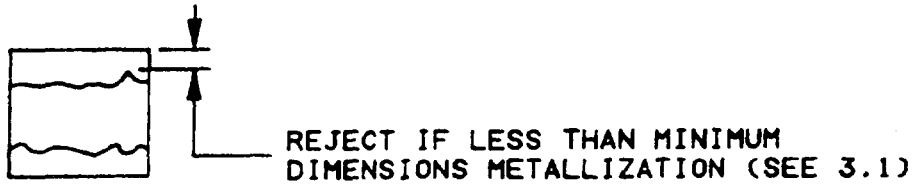


FIGURE C-1. Metallization band gaps.

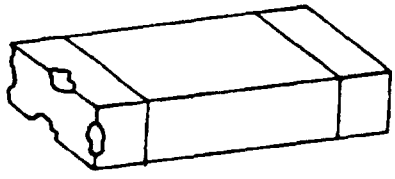


FIGURE C-2. Metallization band chipping.

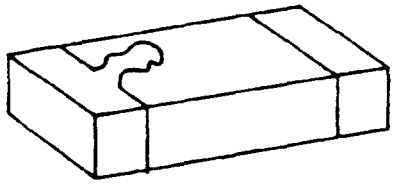


FIGURE C-3. Excessive metallization.

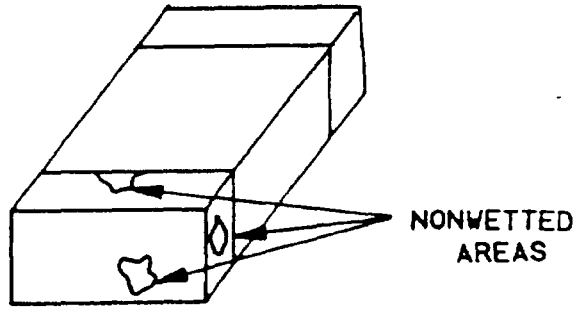


FIGURE C-4. Solder defects.

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30.2.2 Ceramic surface examination.

a. There shall be no cracks (see figure C-5).

b. There shall be no delamination of ceramic layers. A delamination is the separation of ceramic layers (usually observed on sides). (See figure C-6.)

c. Striations are unacceptable (see figure C-6).

d. Edge chip-outs shall not extend to within .003 inch (0.08 mm) of an electrode. An edge chip-out is any region where material has been removed as a result of a fracture (see figure C-7).

e. Rough edges shall be permitted provided they are within the allowable chip-out region as specified on figure C-7.

f. Lips, flared edges, or irregular shapes shall not exceed limits specified on figure C-8.

g. There shall be no fused dust or excess material on external surface that prevents a chip from lying flat, or protrude more than .003 inch (0.08 mm) out of a surface.

h. There shall be no raised surfaces, bubbles, or blisters greater than .002 inch (0.05 mm) (usually found on top and bottom). (See figure C-9.)

i. When compared to a flat surface, the clearance (warpage) at the center of the chip shall be less than 5 percent of the length dimension (see figure C-10).

j. A pinhole is a circular hollow or cavity no larger than .002 inch (0.05 mm) in diameter (see figure C-11). They are unacceptable if they exceed this size.

k. There shall be no holes (voids) that expose electrode plates (see figure C-12).

l. The marking shall be legible and complete (see figure C-13).

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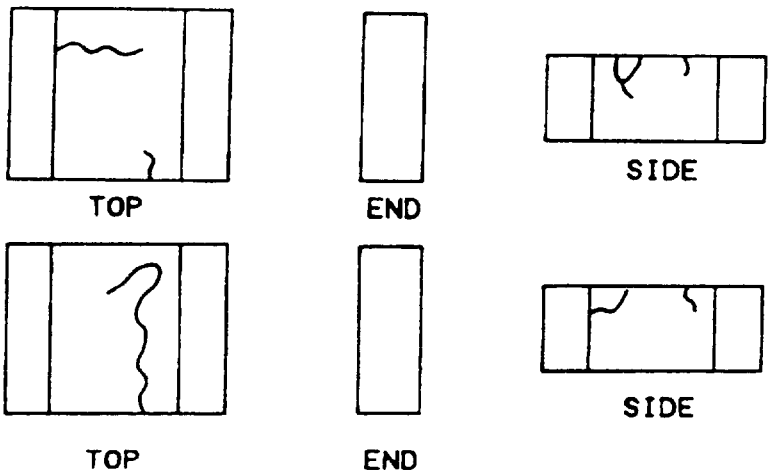


FIGURE C-5. Cracks.

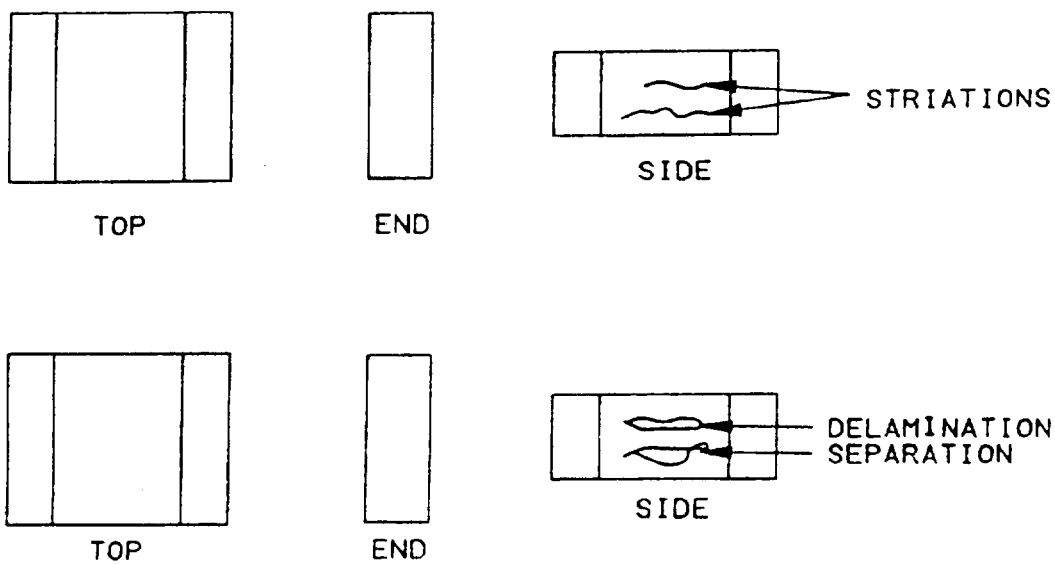


FIGURE C-6. Striations and delaminations.

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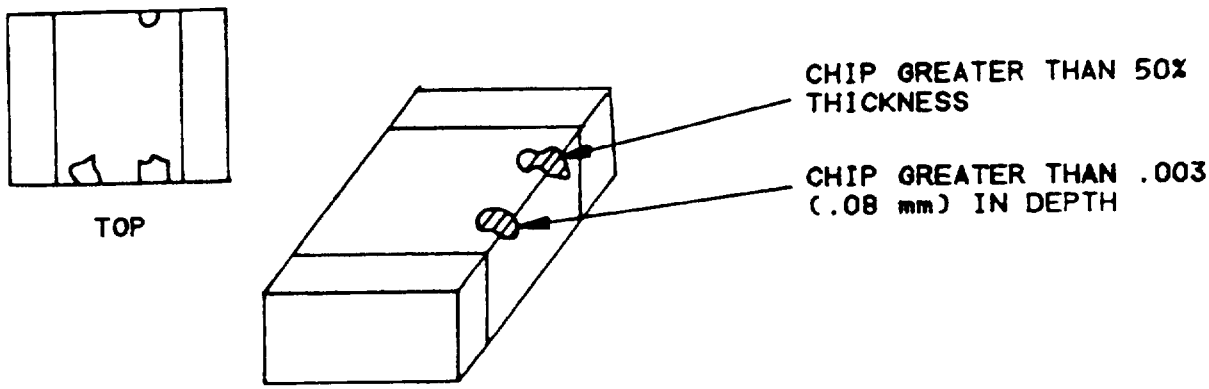


FIGURE C-7. Chip-outs.

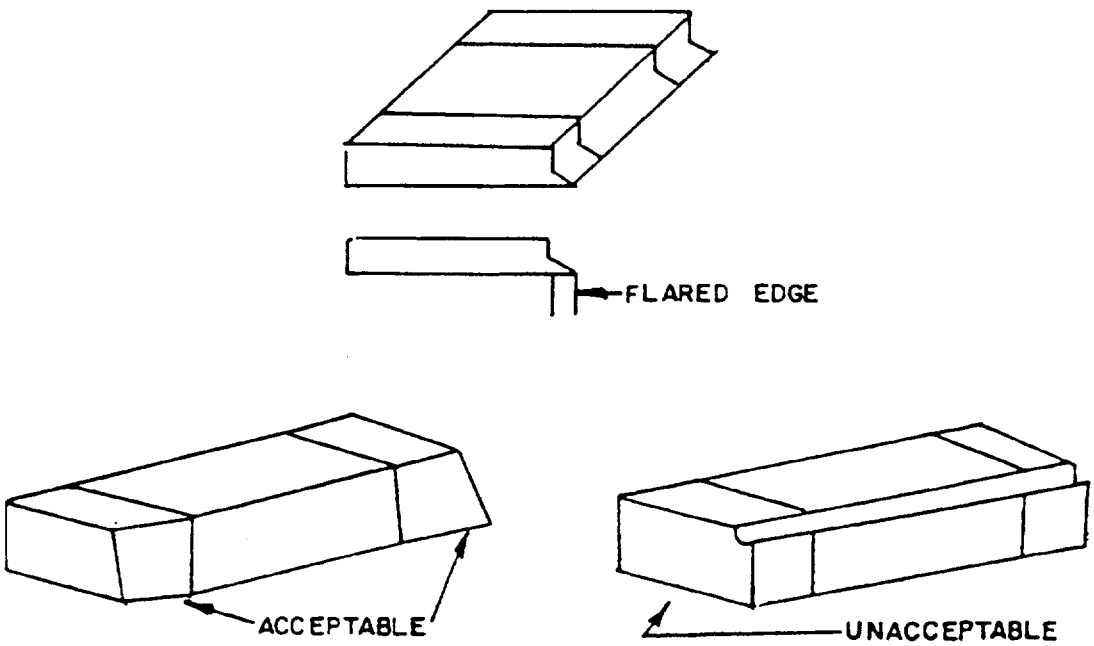


FIGURE C-8. Unacceptable lips, flared edges, or irregular shapes.

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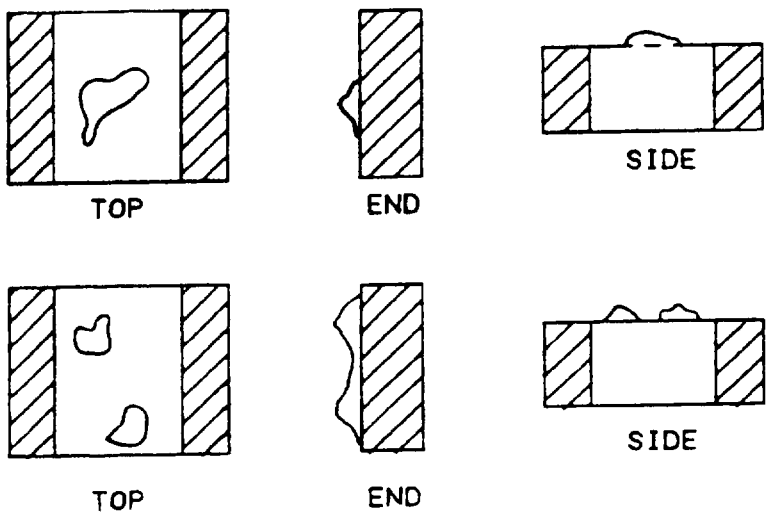


FIGURE C-9. Raised surfaces, bubbles, and blisters.

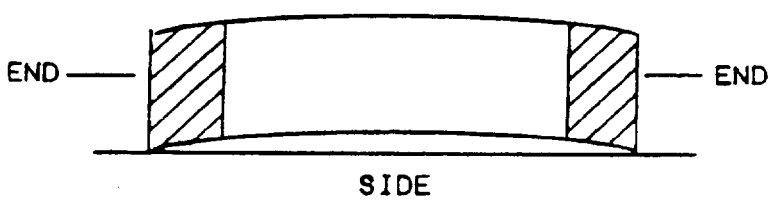


FIGURE C-10. Warpage.

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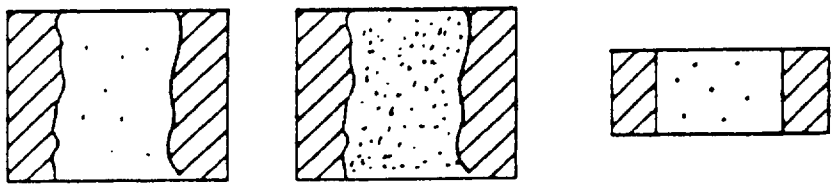


FIGURE C-11. Pinholes.

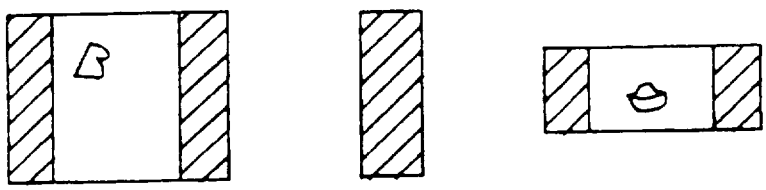


FIGURE C-12. Voids.

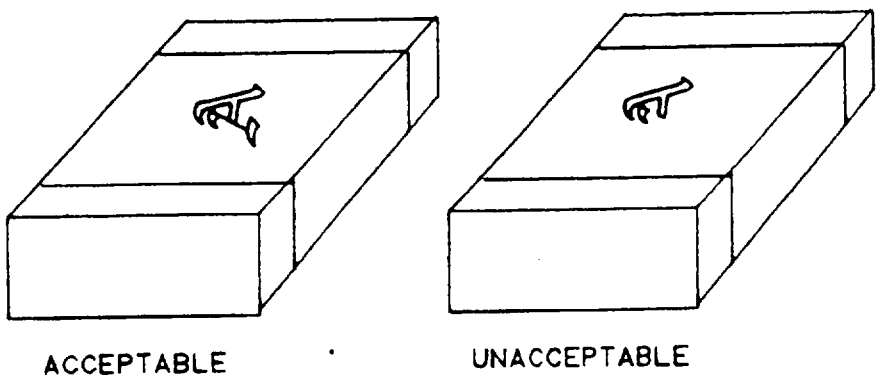


FIGURE C-13. Marking.

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APPENDIX D

LEAD PULL

10. SCOPE

10.1 Scope. This appendix specifies special test procedures and methods for CKS ceramic capacitors. Details for testing of axial and radial leaded, and two-pin dual-in-line package (DIP) capacitors prior to molding or encapsulation is included. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT

30.1 Gauge and test stand. The test equipment used shall be in accordance with the list below or its equivalent.

a. For radial and axial leaded devices:

(1) Chatillon push-pull gauge, model DPP-25.

(2) Chatillon test stand, model LTC, or equivalent.

b. For DIP devices:

(1) Chatillon digital force gauge, model DFG-100.

(2) Chatillon universal test stand, model LTCM II.

(3) Test fixture as shown on figure D-2.

40. TEST METHOD

40.1 Radial devices. Firmly clamp the body of the capacitor into the test stand. Gradually apply an ever-increasing force to one lead at a time in the direction of the leads (see figure D-1) until rupture occurs. Record the force for both leads.

40.2 Axial devices. Firmly clamp one lead into the test stand. Gradually apply an ever-increasing force on the body in the direction of the other lead (see figure D-1) until rupture occurs. Now clamp the body of the device and pull the remaining good lead in the direction of the lead until rupture occurs. Record the stress level for both leads.

40.3 DIP devices.

a. Place sample strip of chips assembled in lead frame on lower fixture so that the paddle portion of the lead frame is up (see figure D-2), and the index hole fits securely around the alignment pin.

b. Gradually apply an ever increasing force to the capacitor body until a rupture occurs. If the capacitor chip breaks during the test, resulting in a rupture force below the minimum allowed, disregard the reading and perform the test on another device.

c. Record the rupture force as indicated on the gauge.

40.4 Minimum rupture force. Minimum rupture force shall be in accordance with table XIV.

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APPENDIX D

50. REJECTION CRITERIA

50.1 Defects. If one failure occurs, additional samples in accordance with table XIII shall be retested with zero failures. If two failures occur, the lot shall be rejected.

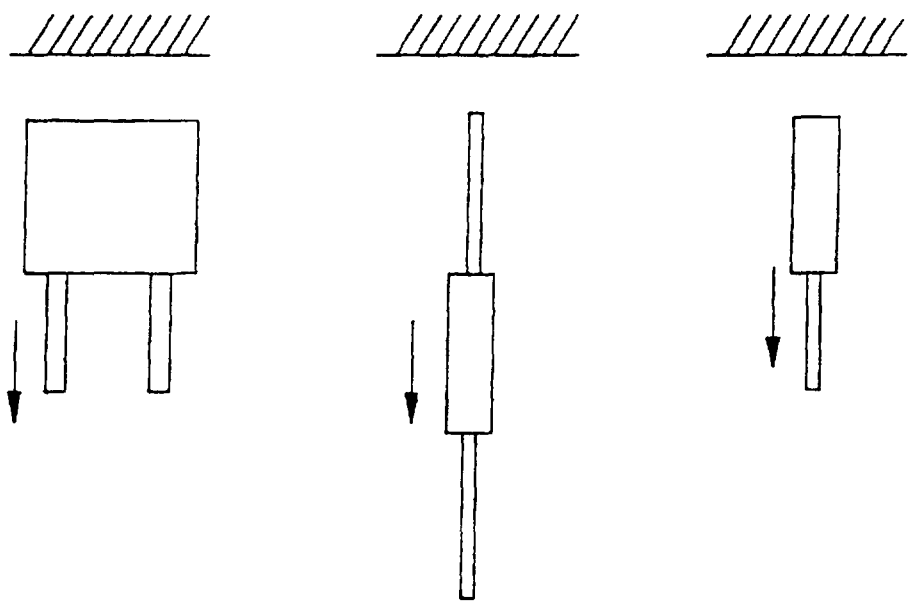
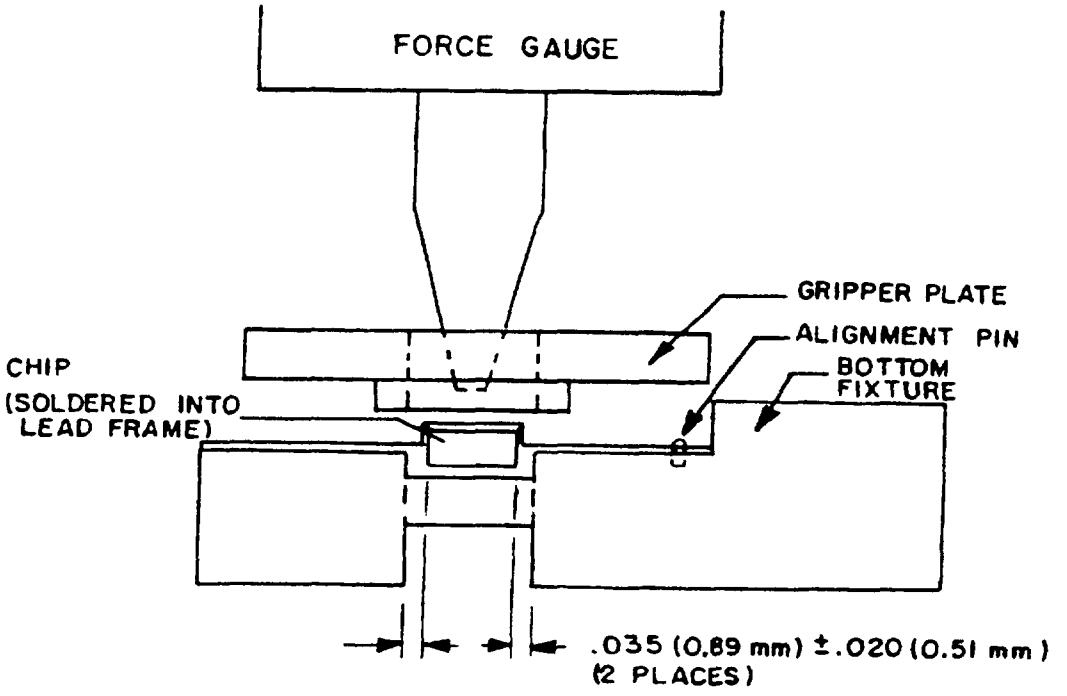


FIGURE D-1. Lead-pull direction.

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NOTE: Force gauge tip diameter = .050 (1.27 mm) ±.010 (0.25 mm).

FIGURE D-2. Text fixture for DIP terminal strength inspection.

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APPENDIX E

RADIOGRAPHIC INSPECTION

10. SCOPE

10.1 Scope. This appendix specifies procedures and practices necessary for radiographic inspection of ceramic capacitors during qualification inspection and group A acceptance inspection. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

10.2 Facilities. Radiographs of the capacitors may be made by the capacitor manufacturer or a suitable independent laboratory. In either case, facilities, equipment, personnel, and techniques shall be in accordance with the requirements specified herein.

10.3 Responsibility for inspection. The capacitor manufacturer shall perform the inspection of the radiograph films and acceptance/rejection of the capacitors.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT

30.1 Dimensions. The dimensional values on which these requirements are based were developed using the U.S. system. The metric equivalents were obtained by direct conversion based upon 1 inch = 25.4 mm. Where such equivalents do not correspond to a standard metric size, the closest standardized metric equivalent size may be used.

30.2 Radiography. The X-ray equipment shall have sufficient voltage range to produce radiographs in accordance with this document. The equipment shall have a focal spot of 3.5 millimeters (mm) or less and shall maintain a sharply defined image at a focal film distance of 30 to 60 inches (76.20 to 152.40 centimeters).

30.3 Exposure factors. The X-ray exposure factors shall be selected to achieve maximum image detail within the sensitivity requirements. The film shall be exposed in accordance with the following requirements:

a. X-ray voltage: Lowest voltage possible.

b. H and D film density: 1.5 to 3.0.

c. Milliampere and time settings: Adjusted, as necessary, to obtain satisfactory exposure.

30.4 Film. The X-ray film shall be single emulsion and of a grade defined as very fine grain.

30.4.1 Sensitivity. X-ray film and equipment shall be capable of detecting metallic particles with a major dimension of .004 inch (0.10 mm) or greater.

30.4.2 Exposure. Exposure factors such as KVP, current, and time shall be compatible with the sensitivity requirements of 30.4.1.

30.4.3 Film density. The X-ray equipment and processing techniques shall be capable of producing H and D film density of 1.5 to 3.0 in accordance with the American Standard Printing density type P-2.

30.4.4 Film dimensions. Radiographic film shall not exceed 14 inches (35.56 centimeters) in width and 17 inches (43.18 centimeters) in length.

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30.4.5 Processing. The exposed X-ray film shall be processed in such manner that the film shall be free of processing defects, i.e., fingerprints, chemical spots, blemishes, etc.

30.4.6 Film identification. Each radiographic film shall be identified with the following information:

a. Part manufacturer's name.

b. Part number (as marked on part).

c. Part cross-reference.

d. Date or lot code (as marked on part).

e. View number.

f. X-ray laboratory name.

g. Penetrameter image.

h. Penetrameter number (see figure E-1).

30.5 Penetrameters. Penetrameters shall be employed in all radiographic testing and shall be as specified in figure E-1. The penetrameter image shall be used to determine radiographic quality and shall meet the following requirements:

a. Penetrameter wires shall be visible on each radiograph.

b. Penetrameters shall be selected to give a film density within ±10 percent of the density of the area of immediate interest.

c. Penetrameters shall be placed in diagonal corners on the source side of the film. The plane of the penetrameters shall be normal to the radiation beam. When 35-mm film strip is used, the penetrameter shall be placed in a position normally occupied by a part, and a penetrameter image shall be made (exposed) for every 50 parts or 17 inches (43.18 centimeters) of film, whichever is more convenient.

d. Distortion of any penetrameter shall not exceed 10 percent.

e. The spacing between wires of a penetrameter shall not be distorted by more than 10 percent. The percentage of distortion as used in this standard is defined as follows:

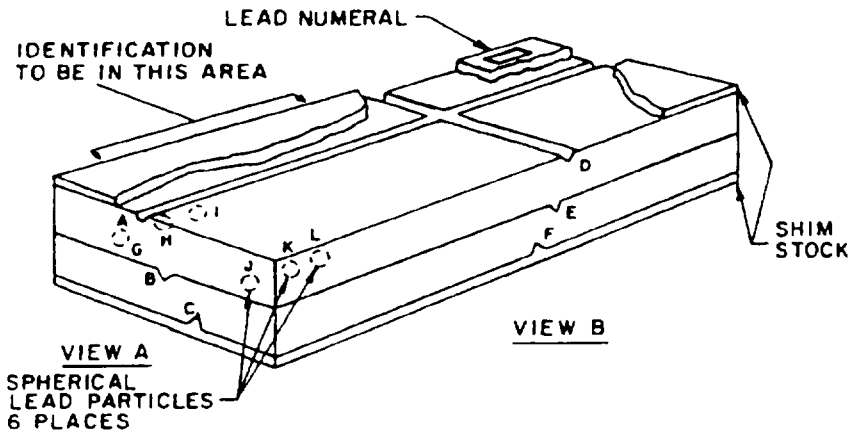
$$\text{Percentage distortion} = \frac{S_0 - S_1}{S_0} \times 100 \text{ where}$$

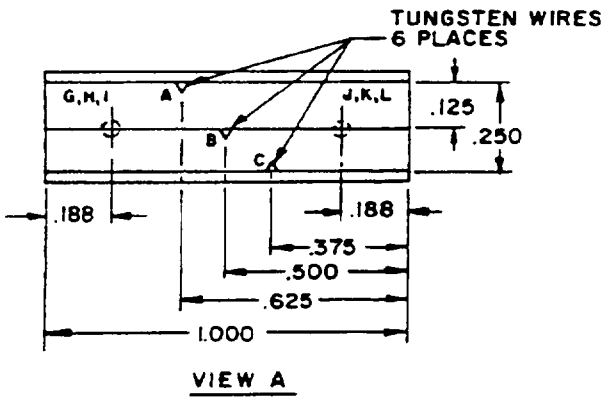
S_0 = actual wire spacing, and
S_1 = wire spacing as it appears on the X-ray film.

30.6 Fixtures. Suitable fixtures shall be used for mounting the electronic parts during the X-ray operation.

30.6.1 Mounting of parts. The capacitors may be mounted in any type of fixture, provided that any metallic portion of the fixture is not between the body of the capacitor and the film.

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Inches	mm	Inches	mm
.125	3.18	.625	15.88
.188	4.78	.750	19.05
.250	6.35	1.000	25.40
.375	9.52	1.500	38.10
.500	12.70	2.250	57.15
.563	14.30		

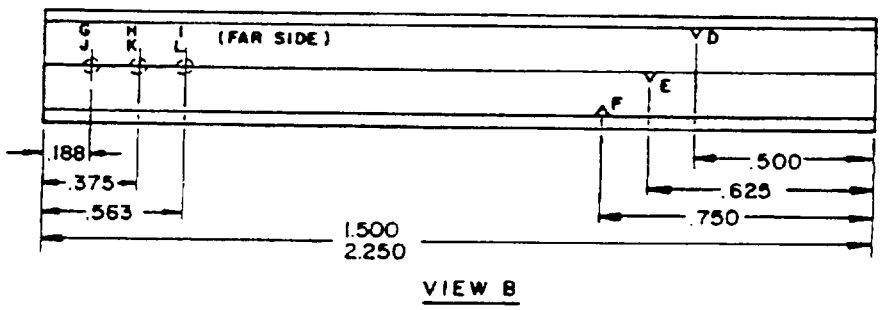


FIGURE E-1. Penetrameter.

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Tungsten wire diameters						Lead particle diameters						Steel shim stock
A	B	C	D	E	F	G	H	I	J	K	L	
.002 (0.05)	.001 (0.03)	.0005 (0.013)	.0005 (0.013)	.001 (0.03)	.002 (0.05)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	None
.002 (0.05)	.001 (0.03)	.0005 (0.013)	.0005 (0.013)	.001 (0.03)	.002 (0.05)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.002 (0.05)
.002 (0.05)	.001 (0.03)	.0005 (0.013)	.0005 (0.013)	.001 (0.03)	.002 (0.05)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.005 (0.13)
.002 (0.05)	.001 (0.03)	.0005 (0.013)	.0005 (0.013)	.001 (0.03)	.002 (0.05)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.007 (0.18)
.003 (0.08)	.002 (0.05)	.001 (0.03)	.001 (0.03)	.002 (0.05)	.003 (0.08)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.010 (0.25)
.003 (0.08)	.002 (0.05)	.001 (0.03)	.001 (0.03)	.002 (0.05)	.003 (0.08)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.015 (0.38)
.005 (0.13)	.003 (0.08)	.002 (0.05)	.002 (0.05)	.003 (0.08)	.005 (0.13)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.025 (0.64)
.005 (0.13)	.003 (0.08)	.002 (0.05)	.002 (0.05)	.003 (0.08)	.005 (0.13)	.015 (0.38)	.010 (0.25)	.008 (0.20)	.006 (0.15)	.004 (0.10)	.002 (0.05)	.035 (0.89)

FIGURE E-1. Penetrameter - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Wires to be tungsten, shim stock to be carbon steel, particles to be lead. Center section to be .125 inch (3.18 mm) layers of clear acrylic plastic, bonded with clear plastic cement of low X-ray density. Fasteners may be used within .250 inch (6.35 mm) or less from each corner, but shall not interfere with end use of the penetrameter. Bottom surface shall be flush.
5. All dimensions shown are ±.005 inch (0.13 mm), except wires, and shim stock, which shall be within standard mil tolerances and lead particles which shall be ±.0002 inch (0.005 mm). Groove details are not critical except that wire must be embedded flush or below surface of plastic and centered at the location shown. Particle-hole sizes are not critical, but should not exceed .031 inch (0.79 mm) in diameter and depth, and must be centered as shown, ±.005 inch (0.13 mm).
6. Additional layers of shim stock may be used as necessary.
7. Identification marking shall be permanent and legible. Location and size of characters are not critical but shall not interfere with or obscure the radiographic image details.

FIGURE E-1. Penetrameter - Continued.

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30.6.2 Identification of capacitors. Each capacitor shall be traceable from its position on the fixture to its corresponding image on the film until completion of film inspection and identification of any with rejectable defects.

30.7 Inspection and examination of radiographs (see figures E-2 and E-3). Inspection of radiographs shall be conducted by the device manufacturer. Each radiograph shall be examined, utilizing the equipment specified herein. The radiographs shall be inspected to determine that each capacitor conforms to the requirements of this document.

30.7.1 Viewing equipment. The radiograph shall be examined on a suitable illuminator with variable intensity, or on a viewer suitable for radiographic inspection of projection-type viewing equipment.

30.7.2 Magnification. A magnification of 10 power minimum shall be used for radiograph examination.

40. REPORTS AND RECORDS

40.1 Report of inspection. Unless otherwise specified by the contract, the testing activity shall furnish inspection reports signed by an authorized representative of the testing activity. The reports shall give the results of the radiographic inspection and shall list the part number, the number of parts inspected, the number of parts rejected, and the date of the test.

40.2 Records of inspection. A complete record of details of inspection shall be kept by the manufacturer or testing laboratory. The record shall list the voltage potentials and currents used in the radiographic process, the time of exposure, the distance of the source of radiation from the surface of the part, the distance of the film from the same surface, the approximate angle between the central beam of radiation and the film, the screens and filters used, the size of the focal spot, the time of development of the film, and the part number and lot or date code of the capacitors.

40.3 Records of radiographs. Each radiograph shall carry a radiograph inspection serial number or code letters to identify the radiograph with the examined parts shown in the radiograph. One complete set of radiographs shall be kept by the manufacturer for a period of 15 years minimum.

50. PERSONNEL

50.1 Radiographer. Personnel engaged in radiographic processing shall be familiar with the requirements of this specification and with all applicable documentation controlling radiographic quality of parts and materials being inspected. They shall be certified by their employer to be capable of producing radiographs which meet the requirements of all applicable documentation.

50.2 Radiographic interpreters. Personnel engaged in the interpretation of radiographs shall be familiar with the requirements of this specification and with all applicable documentation controlling radiographic quality of parts and materials being inspected. They shall be certified by their employer to be capable of evaluating radiographs to determine conformance of parts and materials to the requirements of all applicable documentation.

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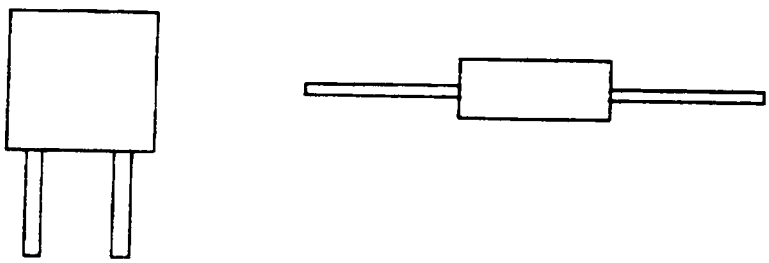
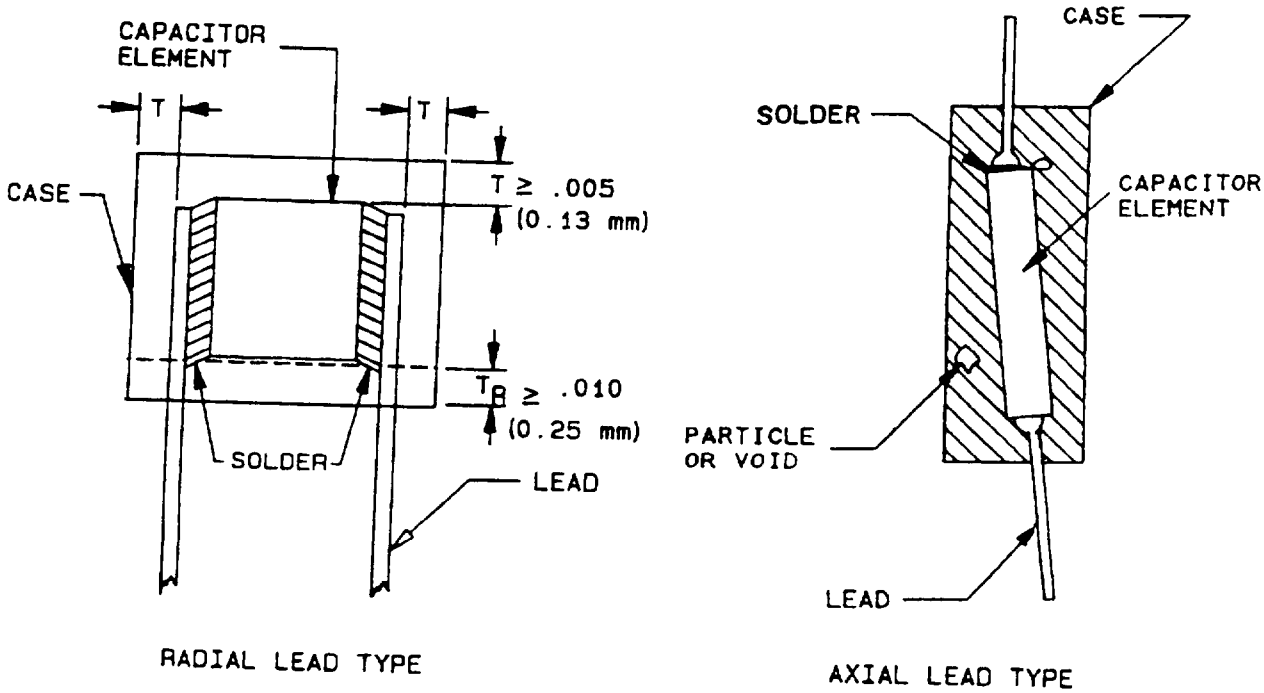


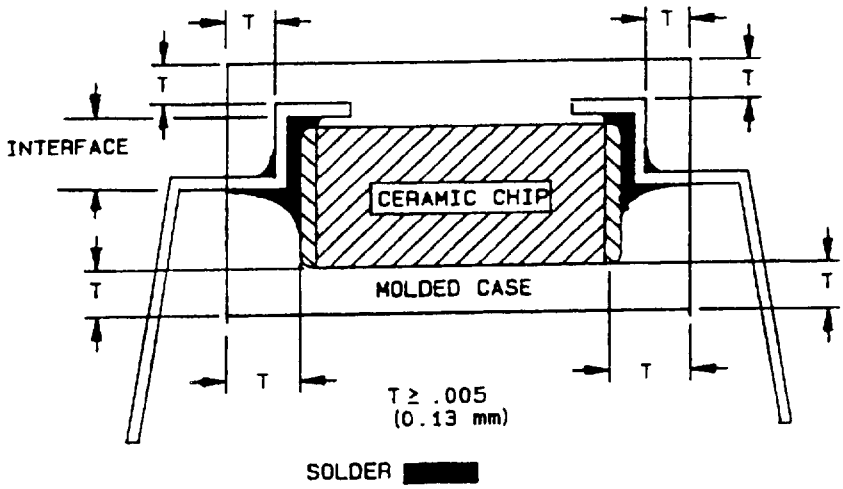
FIGURE E-2. Viewing planes for radiographic inspection.



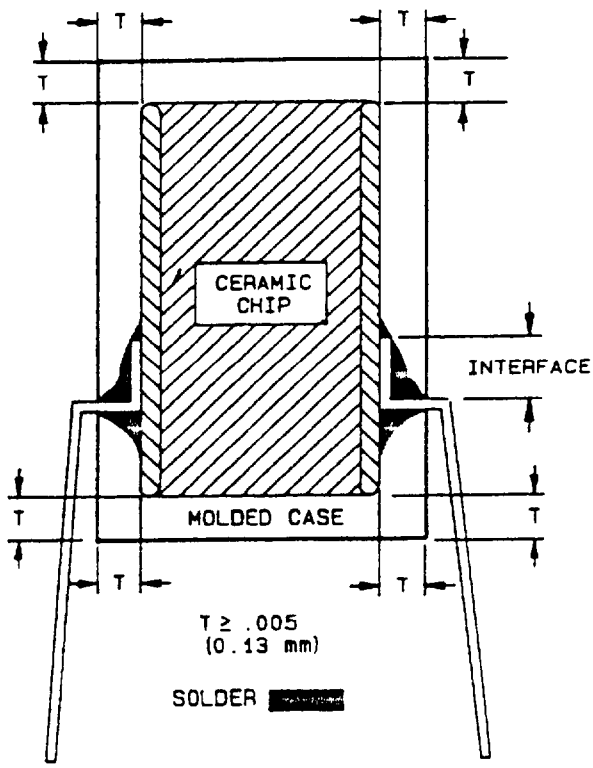
NOTE: T_R is measured to the point at which the lead diameters are increased above their nominal dimensions by the solder meniscus.

FIGURE E-3. Acceptance criteria for radiographic inspection.

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Styles CKS22, CKS23, and CKS24



Style CKS24 alternate

FIGURE E-3. Acceptance criteria for radiographic inspection - Continued.

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50.3 Vision. The minimum vision requirements for visual acuity of personnel inspecting film shall be as follows:

a. Distant vision shall equal 20/30 in at least one eye, either corrected or uncorrected.

b. Near vision shall be such that the individual can read Jaeger type no. 2 at a distance of 16 inches (40.64 centimeters) either corrected or uncorrected.

50.4 Vision tests. Vision tests shall be performed by an oculist, optometrist, or by other professionally recognized personnel. One year from the effective due date of original certification, and each year thereafter, certified personnel shall be required to pass the vision tests specified herein.

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APPENDIX F

EQUIVALENT SERIES RESISTANCE (ESR) AND RESONANCE FREQUENCY MEASUREMENT CRITERIA FOR HIGH FREQUENCY NONLEADED CAPACITOR STYLES CKS28 AND CKS29

10. SCOPE

10.1 Scope. This appendix specifies the method to determine high frequency loss as measured by equivalent series resistance and the maximum use frequency as determined by parallel resonance of chip mounted with plane of electrodes perpendicular to plane of substrate. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Other publications. The following document forms a part of the specification to the extent specified herein.

ELECTRONIC INDUSTRIES ASSOCIATION

RS-483 - Test for Effective Series Resistance.

(Application for copies should be addressed to the Electronic Industries Association (EIA), 1722 Eye Street, N.W., Washington, DC 20006.)

30. PROCEDURES FOR TEST

30.1 Equivalent series resistance. When specified, the equivalent series resistance shall be measured in accordance with EIA RS-483.

30.2 Resonance frequency.

a. When specified, this measurement is made on the nonleaded capacitor mounted on microstripline in a series connection to the network analyzer.

b. Frequency: Measurement frequency shall be varied smoothly from 100 MHz to 10,000 MHz.

c. Mounting fixture: The test fixture is a 50-ohm microstripline with a gap suitable for mounting unleaded capacitors in series with the line. The fixture shall not resonate within the test frequency specified in 30.2b or the measurements shall be compensated by subtracting a through line calibration measurement.

d. Mounting: The unleaded capacitor is mounted with electrode plane parallel to the microstripline plane.

e. Resonance frequency is the center frequency of the lowest observed resonance.

40. TEST CRITERIA

40.1 Equivalent series resistance. When equivalent series resistance is tested in accordance with 30.1, ESR shall be less than the limits shown on figures F-1 and F-2 at the specified measurement frequency range.

40.2 Resonance frequency. When resonance frequency is tested in accordance with 30.2, the resonance frequency shall meet or exceed the resonance frequency shown on figure F-3.

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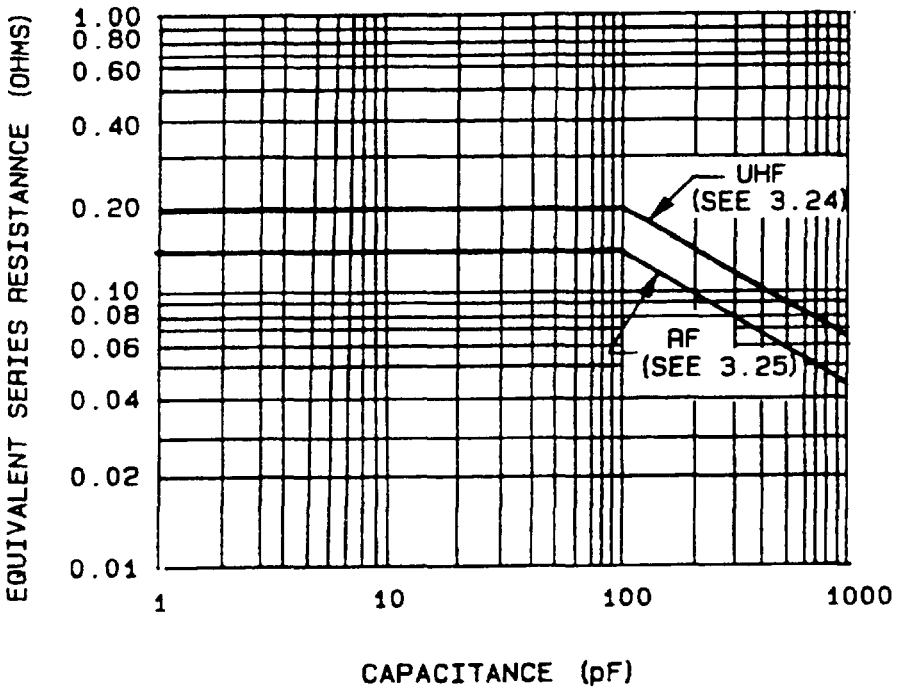


FIGURE F-1. Equivalent series resistance (BG characteristic).

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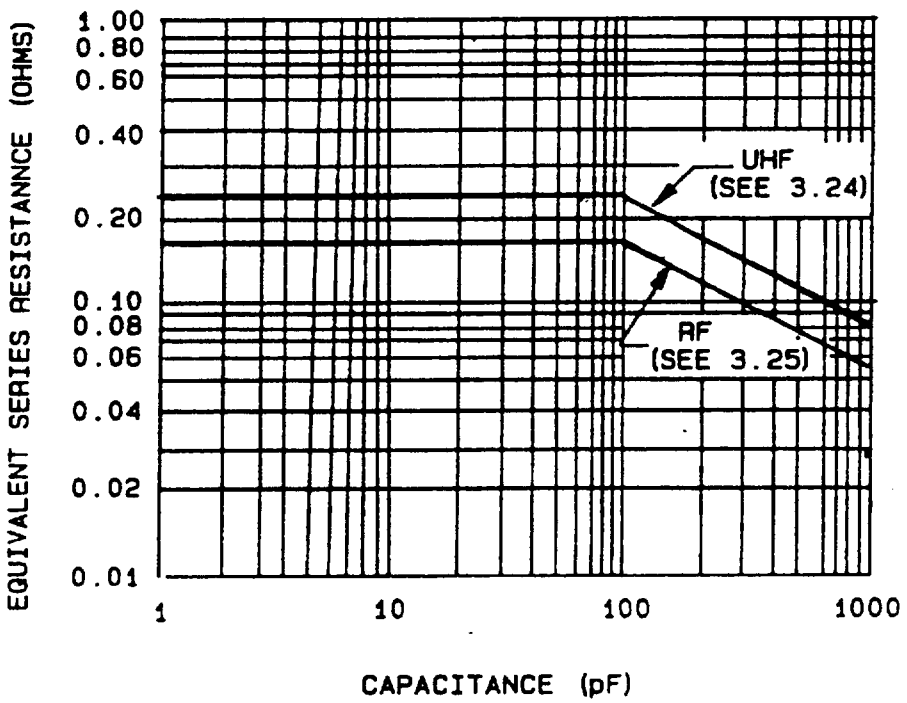


FIGURE F-2. Equivalent series resistance (BP characteristic).

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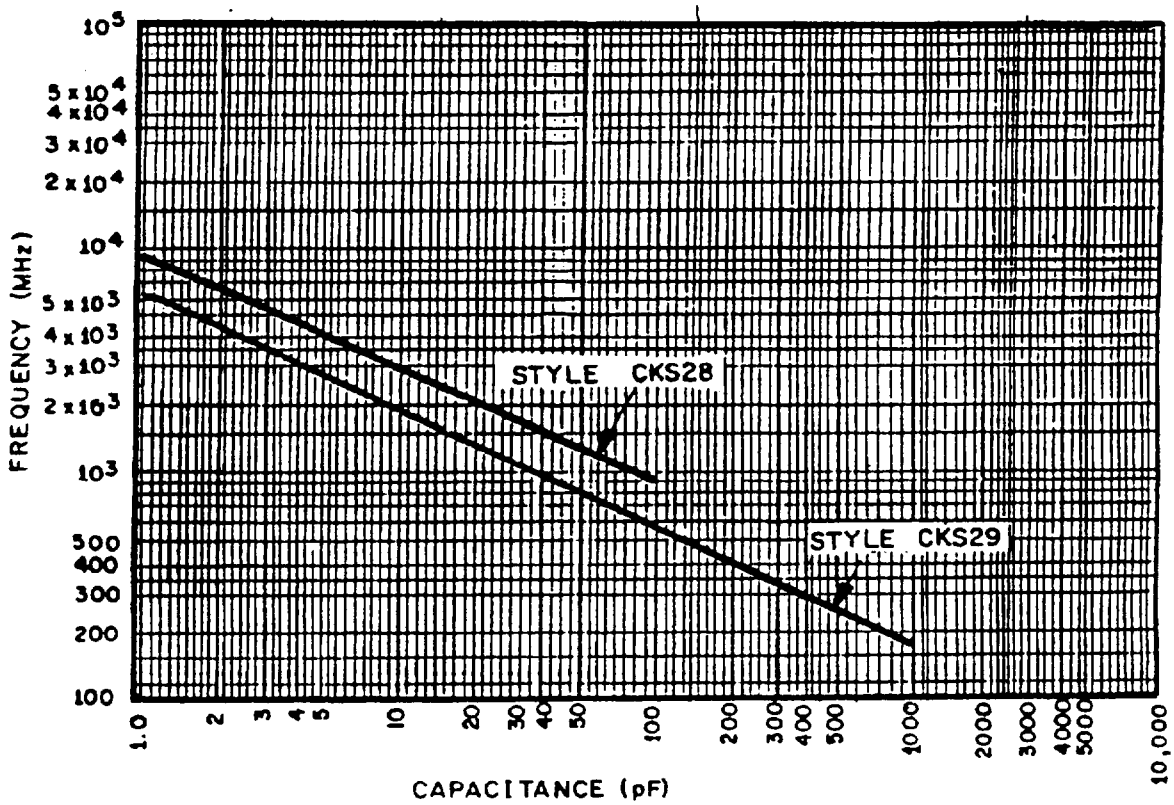


FIGURE F-3. Resonance frequency.

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APPENDIX G

PROCEDURES FOR THE PRODUCT AND QUALITY AUDIT

10. SCOPE

10.1 Scope. This appendix provides procedures to the Product and Quality Audit team for the auditing of manufacturers supplying to this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. INSTRUCTIONS

30.1 Baseline document. The manufacturer shall complete the baseline document and submit the required information to the qualifying activity prior to scheduling the Product and Quality Audit. It is not intended that the manufacturer submit proprietary information. Each manufacturer shall provide a detailed process flow chart which shall include, as a minimum:

a. All major process steps and inspection points by descriptive title, document number, and revision.

b. The identification of all critical manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline approval.

30.1.1 Changes to baseline. The audit team shall compare the latest baseline document to the previous baseline document to assure all changes have been submitted to the qualifying activity and have been approved.

a. Minor changes: Any subsequent departure from the approved flow chart (i.e., additions or deletions), or changes to the revision status of those processes or inspections are allowed, provided they do not violate the changes prohibited by Section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes: No changes (i.e., substitutions) shall be allowed in the following areas without prior review and approval of the qualifying activity. In the following areas, partial or total requalification may be required as determined by the qualifying activity.

(1) Dielectric formulation.

(2) Electrode formulation.

(3) End termination formulation.

(4) Solder formulation.

(5) Lead wire material.

(6) Encapsulation material.

(7) Capacitor design (i.e., dielectric thickness, number of electrodes, chip size, except as permitted by the baseline).

(8) Basic buildup system.

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30.2 Audits. An initial audit and subsequent periodic audits (every year or as deemed necessary) shall be performed by the qualifying activity. The audit shall include, but not be limited to, all of the items on the checklist. The results of any additional items that are assessed shall be added to the checklist for record purposes.

30.2.1 Duties of the audit team.

a. The audit team shall review the baseline document for adequacy and completeness prior to the audit of the manufacturer and should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that all measuring and test equipment is functioning and calibrated. They shall also verify that test measuring procedures and equipment are sufficiently accurate for the required test.

c. The audit team shall typically sample check 12 months of records for the item being certified. In addition, the location of the equipment and the adequacy of the operating procedure shall be checked and, whenever practical, operations utilizing the equipment and procedures witnessed.

d. The audit team shall insure that adequate documentation, trained personnel, equipment, and visual aids are available to perform the inspections specified in table G-I.

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APPENDIX G

TABLE G-I. Certification checklist.

Inspection	Inspector	Satisfactory	Unsatisfactory
a. Verification of MIL-STD-790 requirements.			
b. Flow chart			
(1) Key processes identified with revision.			
(2) Key inspection identified with revision.			
(3) Non-proprietary documents reviewed.			
(4) Screens/inspection points identified.			
c. Baseline - shall describe as a minimum the following:			
(1) Capacitor design by part number indicating the following with the necessary tolerance defined:			
(a) Dielectric type.			
(b) Dielectric thickness.			
(c) Number of electrodes.			
(d) Chip size.			
(2) Basic electrode formulation identification.			
(3) Basic end termination formulation identification.			
(4) Solder formulation identification.			
(5) Lead wire.			
(6) Encapsulation material identification.			
(7) Internal visual prior to encapsulation.			
d. Lot control plan (see 4.5.2.1).			
e. Traceability plan.			
f. Training plan for QC personnel.			
g. Nondestructive test plan.			

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APPENDIX H

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein. There shall be no combining of styles, operating voltage, or characteristic. Each shall be qualified separately. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. SUBMISSION

30.1 Qualification based on similar established reliability (ER) styles. The manufacturer seeking qualification for a CKS style who is qualified to "R" failure rate level of a similar ER style (see table H-I) shall submit to test the total samples specified in table VIII, divided equally into groups consisting of the highest capacitance value for each combination of voltage rating and characteristic.

TABLE H-I. Similar ER styles.

CKS style	Similar style	CKS style	Similar style	CKS style	Similar style
CKS05-BP	CCR05	CKS12-BP	CCR76	CKS24	CKR24
CKS05-BX	CKR05	CKS12-BX	CKR12	CKS26	CKR06
CKS05S-BP	CCR05	CKS14-BP	CCR77	CKS28-BG	CDR11
CKS05S-BX	CKR05	CKS14-BX	CKR14	CKS28-BP	CDR12
CKS06-BP	CCR06	CKS15-BP	CCR78	CKS29-BG	CDR13
CKS06-BX	CKR06	CKS15-BX	CKR15	CKS29-BP	CDR14
CKS06S-BP	CCR06	CKS16-BP	CCR79	CKS51	CDR01
CKS06S-BX	CKR06	CKS16-BX	CKR16	CKS52	CDR02
CKS07-BP	CCR07	CKS22-BP	CKR22	CKS53	CKR03
CKS07-BX	---	CKS22-BX	CKR22	CKS54	CDR06
CKS11-BP	CCR75	CKS23-BP	CKR23		
CKS11-BX	CKR11	CKS23-BX	CKR23		

30.2 Qualification not based on similar established reliability styles. A sample of the size required in table VIII, of the highest capacitance value in each voltage rating and characteristic in each style for which qualification is sought shall be submitted. For example, style CKS05 shall require 752 total sample units to qualify for the entire style.

30.3 Test data. When examinations and tests are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the examinations and tests indicated as nondestructive in table VIII. Each submission shall be accompanied by the test data obtained from these examinations and tests. The performance of the destructive tests by the supplier on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

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30.4 Certification of material. When submitting samples for qualification, the contractor shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements and the baseline documentation.

30.5 Description of items. The manufacturer shall submit a detailed description of the capacitors being submitted for inspection, including body, electrode material, terminal leads, etc.

40. EXTENT OF QUALIFICATION

40.1 Restrictions on qualification. Capacitance-range qualification shall be restricted to values equal to and less than the capacitance value submitted. Capacitance-tolerance qualification will be restricted to tolerances equal to and wider than the tolerance submitted. DC rated voltage, operating temperature range, and voltage-temperature limit qualification shall be restricted to that submitted.

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CONCLUDING MATERIAL

Custodians:
Air Force - 19
NASA - NA

Review activities:
Army - ER
Navy - EC
Air Force - 85
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5910-1608)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-123B	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
Capacitors, Fixed, Ceramic Dielectric, (Temperature Stable and General Purpose), High Relia

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME NASA Parts Project Office Code 311.A (G. Kramer) Goddard Space Flight Center	b. TELEPHONE (Include Area Code) (1) Commercial (301) 286-8884 (2) AUTOVON
c. ADDRESS (Include Zip Code) Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete. — *98/29

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